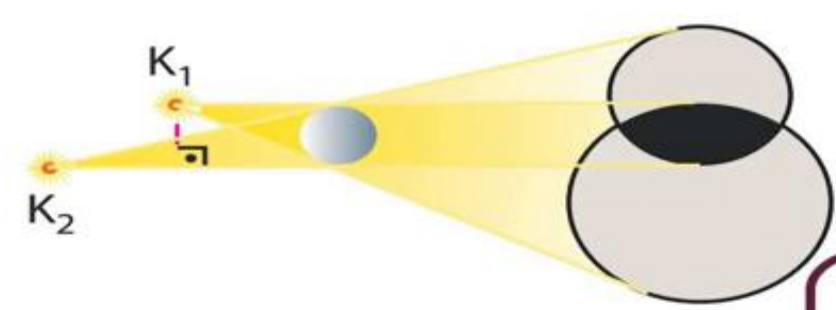


CHAPTER 14 : RAY OPTICS

Learning Objectives :

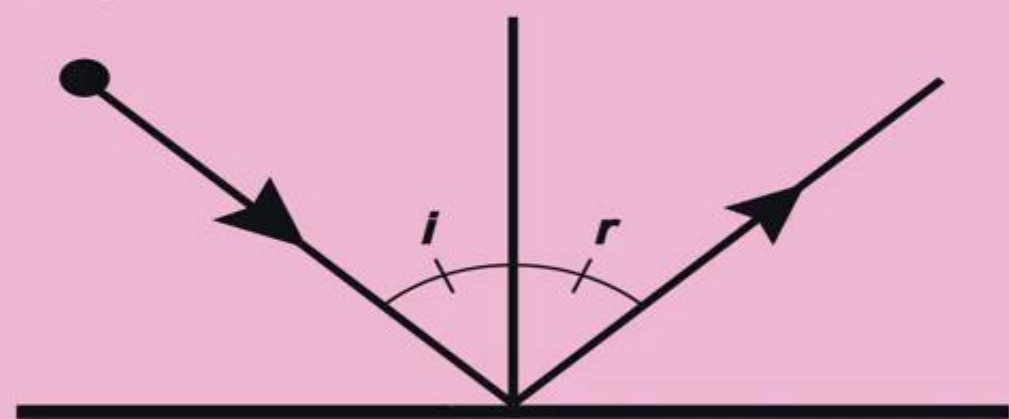
- Reflection of light, spherical mirrors, mirror formula.)
- Refraction of light, total internal reflection and optical fibers.
- Refraction at spherical surfaces, lenses, thin lens formula, lens maker's formula.
- Magnification, power of a lens, combination of thin lenses in contact.
- Refraction of light through a prism.
- Optical instruments: Microscopes and Astronomical telescopes (reflecting and refracting) and their magnifying powers.



Ray Optics

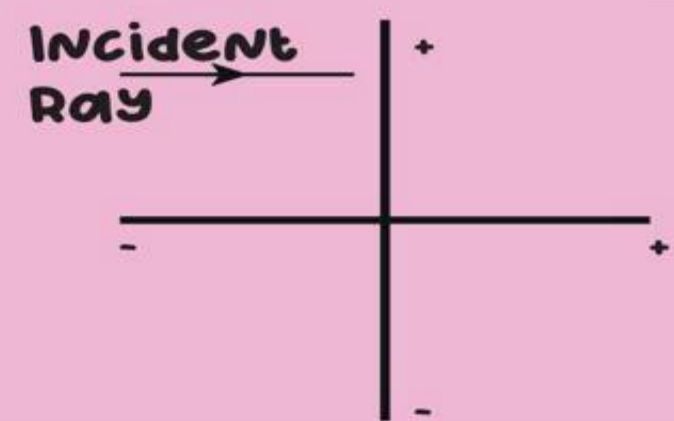
Laws of Reflection

1. The incident ray reflected ray and normal to the reflecting surface all lie in same plane.
2. Angle of reflection is always equal to angle of incidence, i.e., $\angle i = \angle r$



Sign Convention

1. All distances are measured from the pole and is the origin.
2. Distances measured to the right of the pole are taken as positive.
3. Distance above the principal axis are taken as positive.
4. Angle measured from the normal in the anti-clockwise direction are positive.



Absolute Refractive Index

It is defined as ratio of speed of light in vacuum to speed of light in medium

$$\text{i.e. } n = \frac{c}{v}$$

Laws of Refraction

1. The incident ray, refracted ray and normal to the interface of two media all lie on the same plane.
2. Snell's law $\mu_2 \sin r = \mu_1 \sin i$

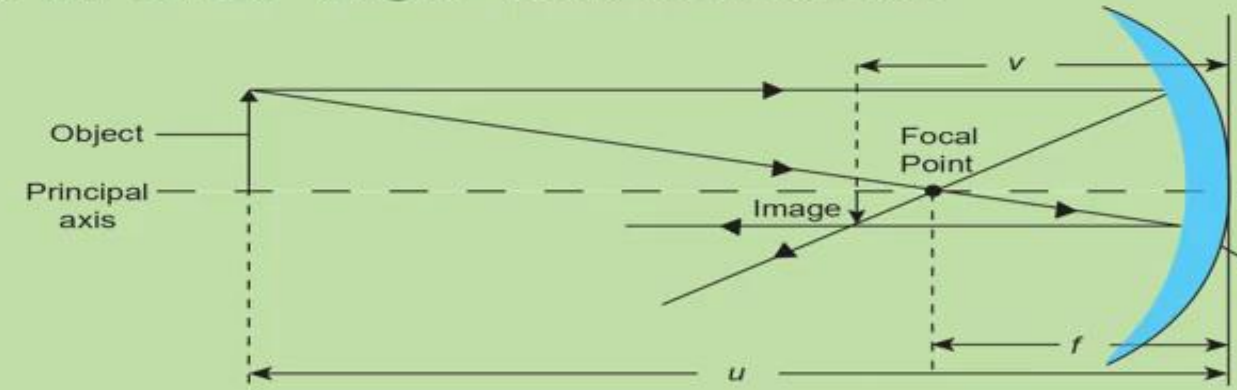


Spherical Mirrors

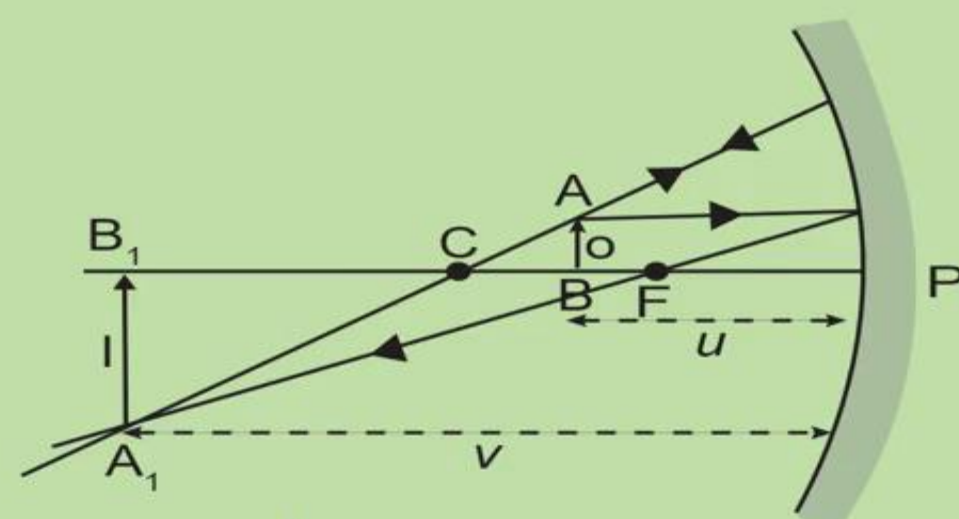
Mirror formula

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

IN PROPER SIGN CONVENTION.



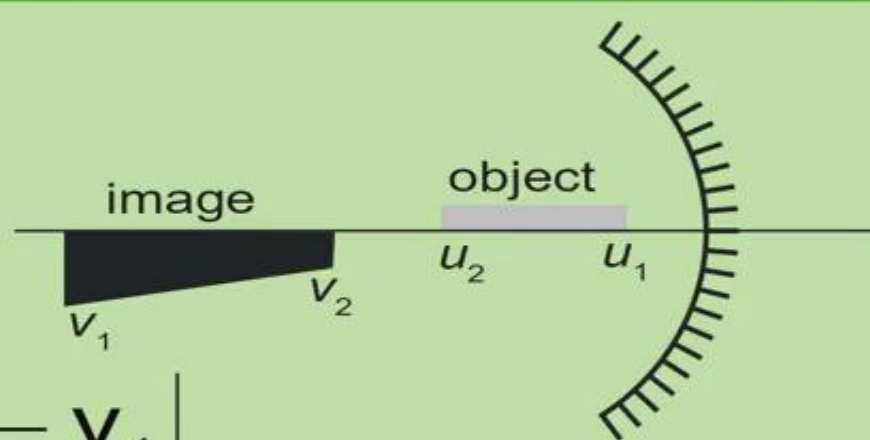
Linear magnification



$$m = \frac{I}{O} = \frac{-v}{u}$$

IN PROPER SIGN CONVENTION

Longitudinal magnification

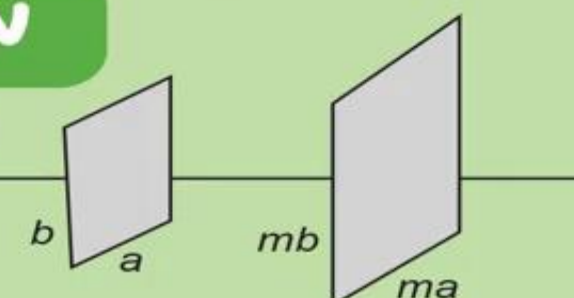


$$m_L = \left| \frac{v_2 - v_1}{u_2 - u_1} \right|$$

For small objects, $m_L = m^2$

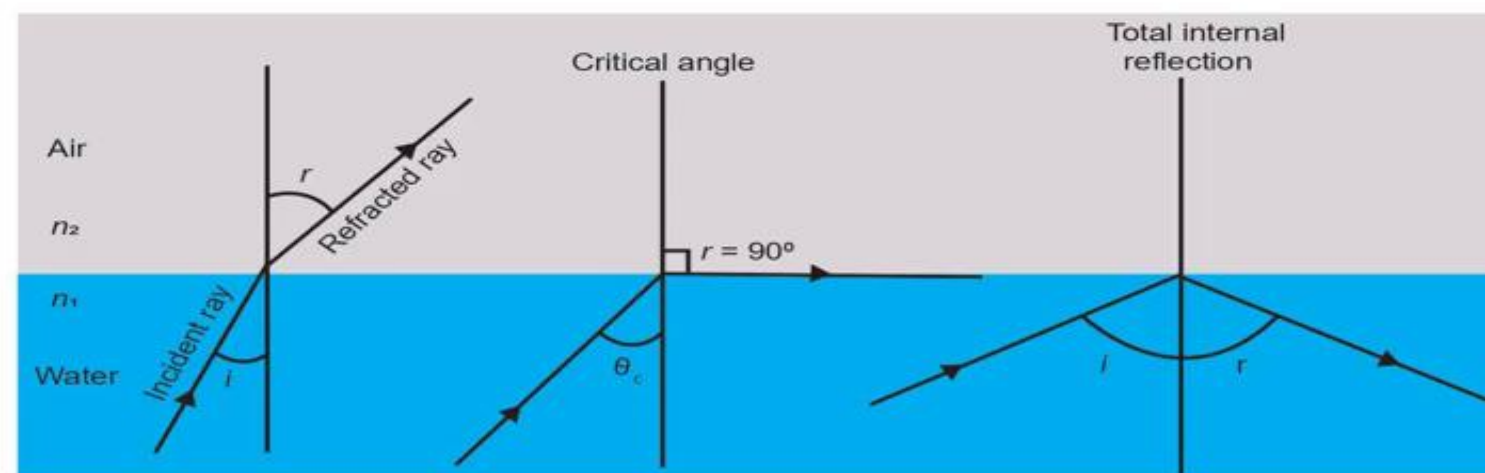
Superficial Magnification

$$m_s = \frac{m_a \times m_b}{a \times b} \quad \text{For small objects, } m_L = m^2$$
$$= m^2$$



TIR

The balancing back of light ray in the same denser medium after reflection from an interface with a rarer medium is termed as total internal reflection.



Critical Angle

It is the angle of incidence for which the angle of refraction is 90°.

$$\theta_c = \sin^{-1} \left(\frac{n_2}{n_1} \right)$$

Conditions for TIR

1. The light ray must travel from denser to rarer medium.
2. The angle of incidence must be greater than the critical angle.

Application of TIR

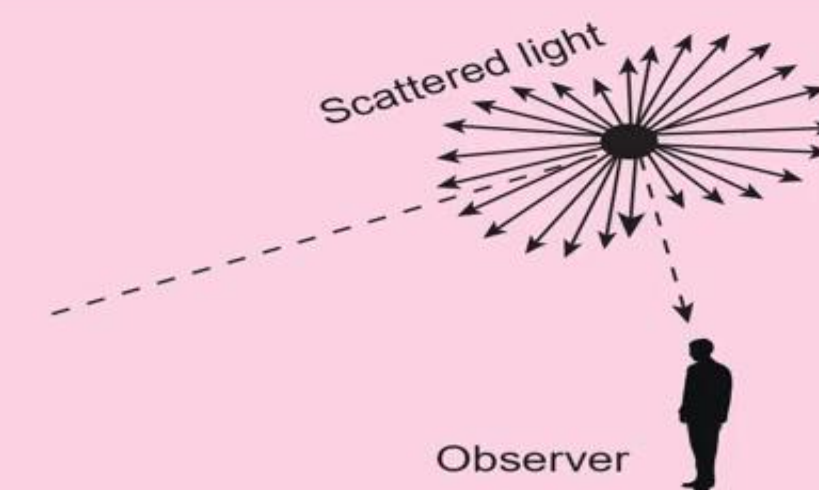
1. Sparkling of diamond
2. Optical fibre
3. Mirage and optical looming.

Scattering

The deflection of light ray by the fine particles of matter is known as scattering of light. From Rayleigh scattering.

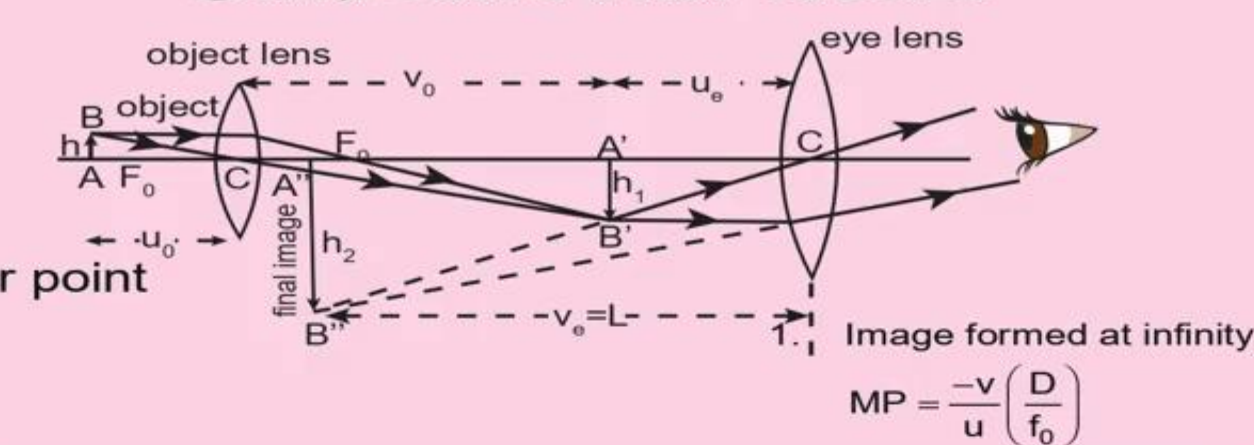
$$I \propto \frac{1}{\lambda^4}$$

where λ is wavelength of light and I is intensity of light.



Optical Instrument

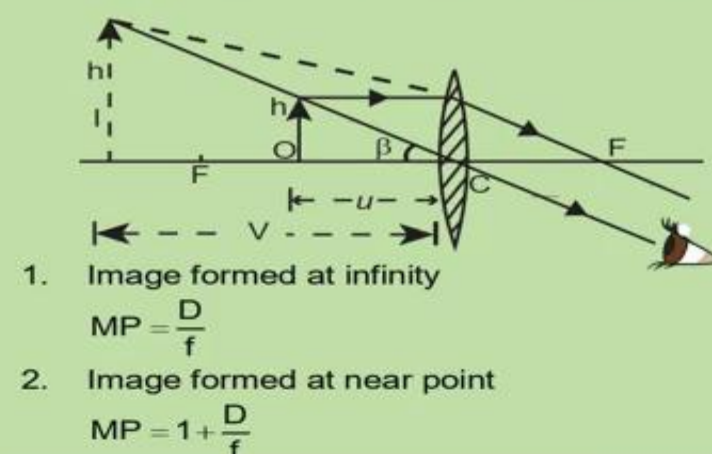
Compound Microscope



2. Image formed at near point
- $$cMP = \frac{-v}{u} \left(1 + \frac{D}{f_0} \right)$$

$$MP = \frac{-v}{u} \left(\frac{D}{f_0} \right)$$

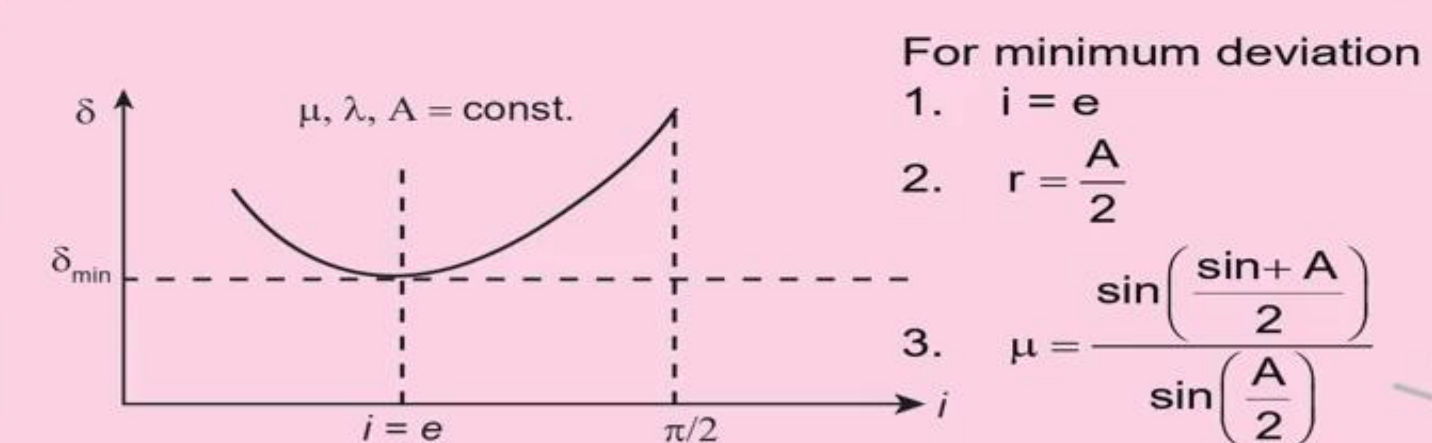
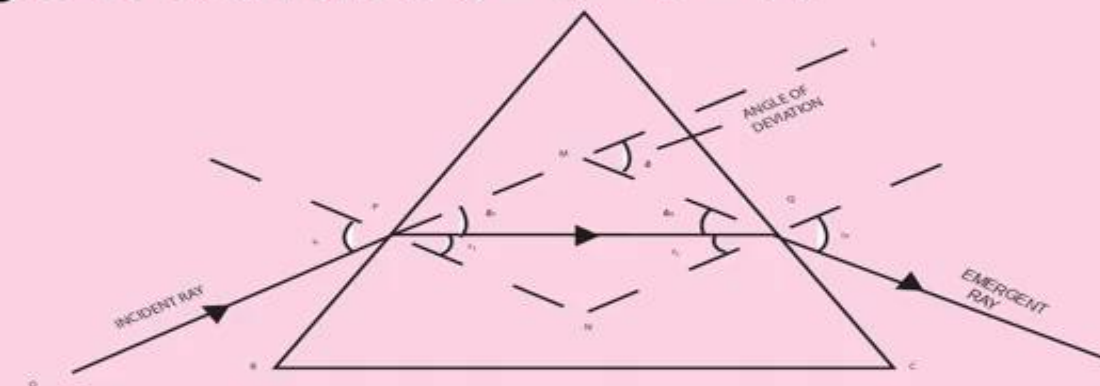
Simple Microscope



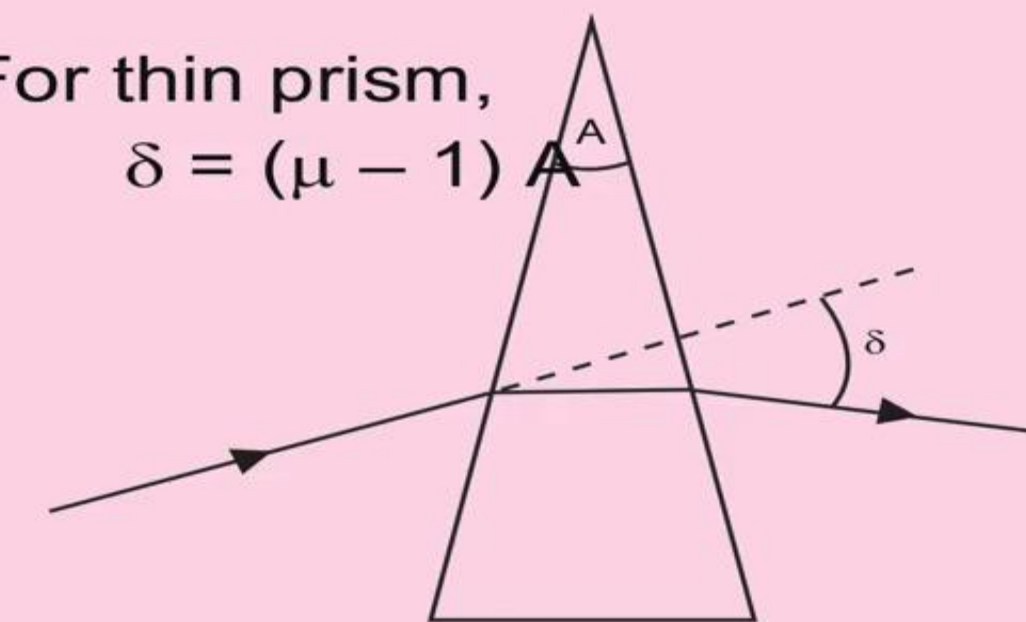
1. Image formed at infinity
 2. Image formed at near point
- $$MP = \frac{D}{f}$$
- $$MP = 1 + \frac{D}{f}$$

Prism

Angle of deviation $\delta = i + e - A$

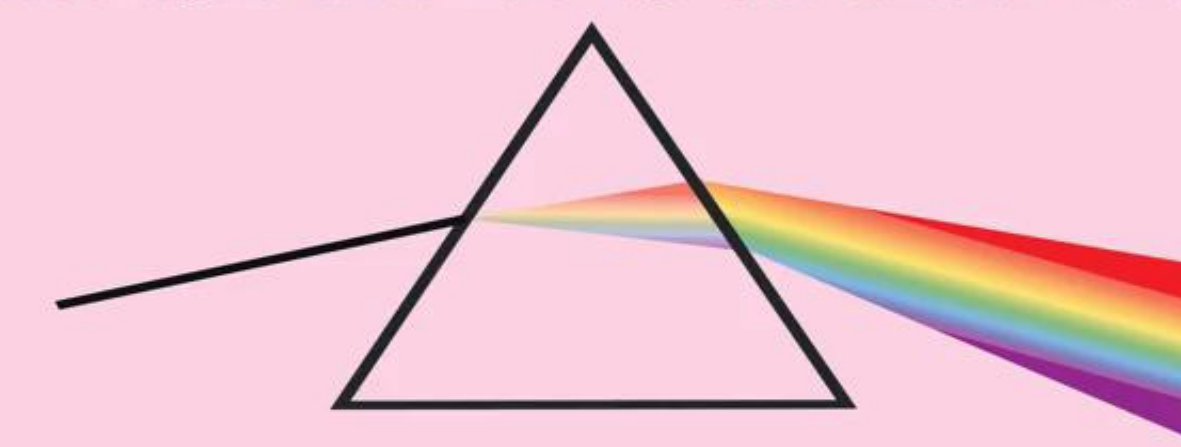


For thin prism,
 $\delta = (\mu - 1) A$

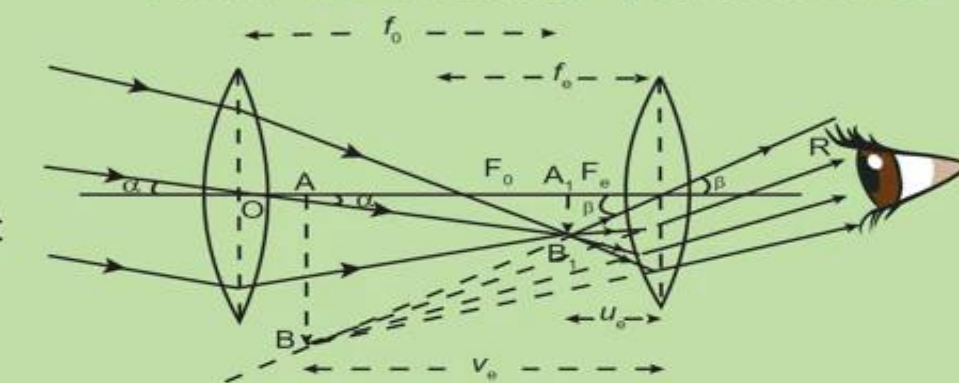


Dispersion through Prism

When white light passes through the prism, then it splits into its seven constituent colours. This phenomenon of splitting of white light is known as dispersion of light.

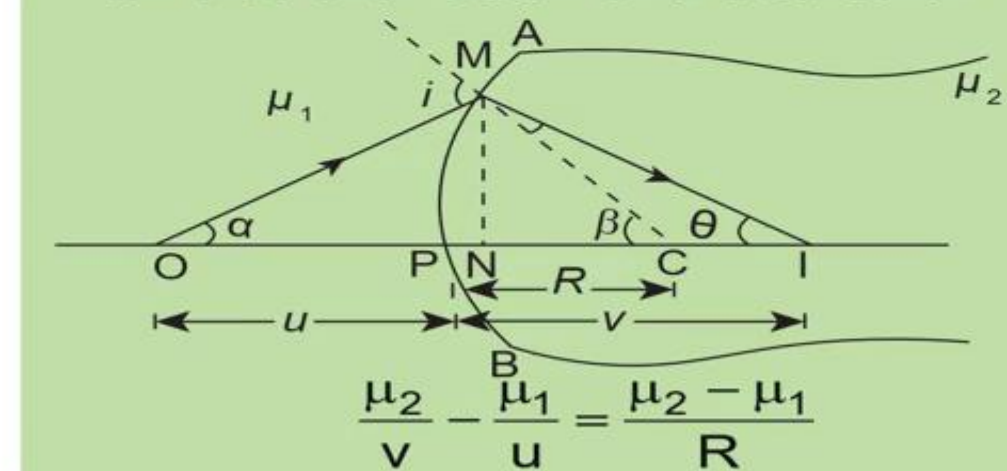


Astronomical Telescope

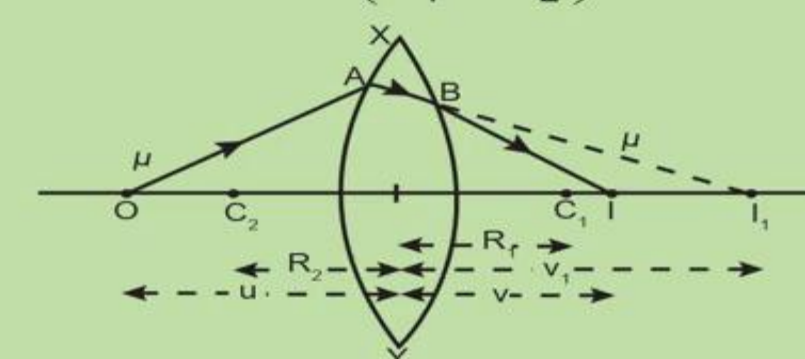


1. Image formed at infinity
 2. Image formed at near point
- $$MP = \frac{-f_0}{u_0}$$
- $$MP = \frac{-f_0}{f_e} \left[1 + \frac{f_e}{D} \right]$$

IN PERSON SIGN CONVENTION

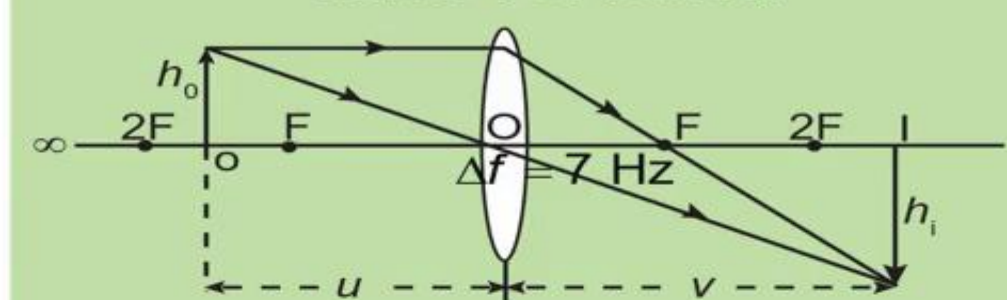


$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$



IN PROPER SIGN CONVENTION

LENS FORMULA



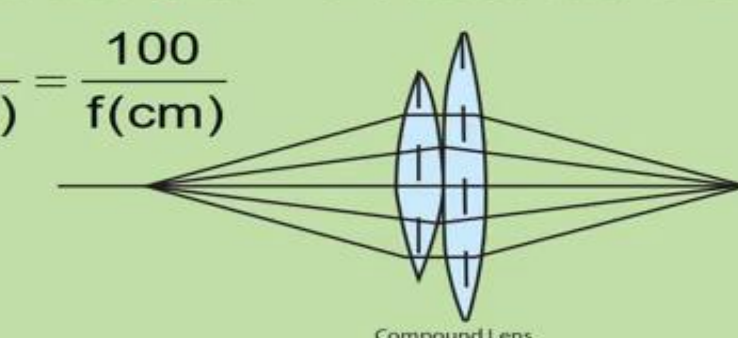
$$\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$$
$$m = \frac{h'}{h} = \frac{v}{u}$$

IN PERSON SIGN CONVENTION

Power of lens

It is defined as the reciprocal of focal length of metres, i.e.,

$$P = \frac{1}{f(m)} = \frac{100}{f(cm)}$$



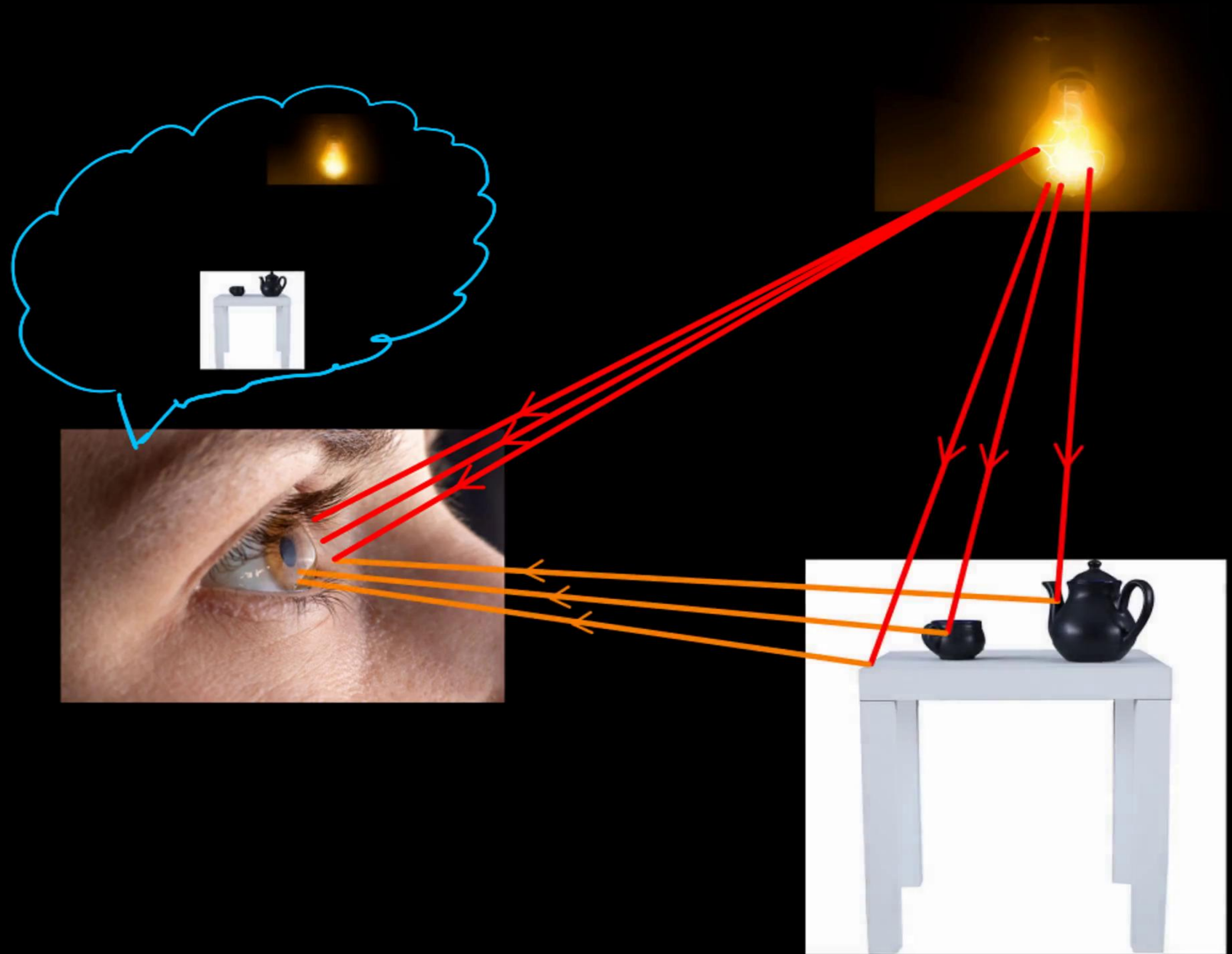
For combination of lenses,
 $P = P_1 + P_2$

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2}$$

In proper sign convention

Introduction

What makes things visible?



Introduction

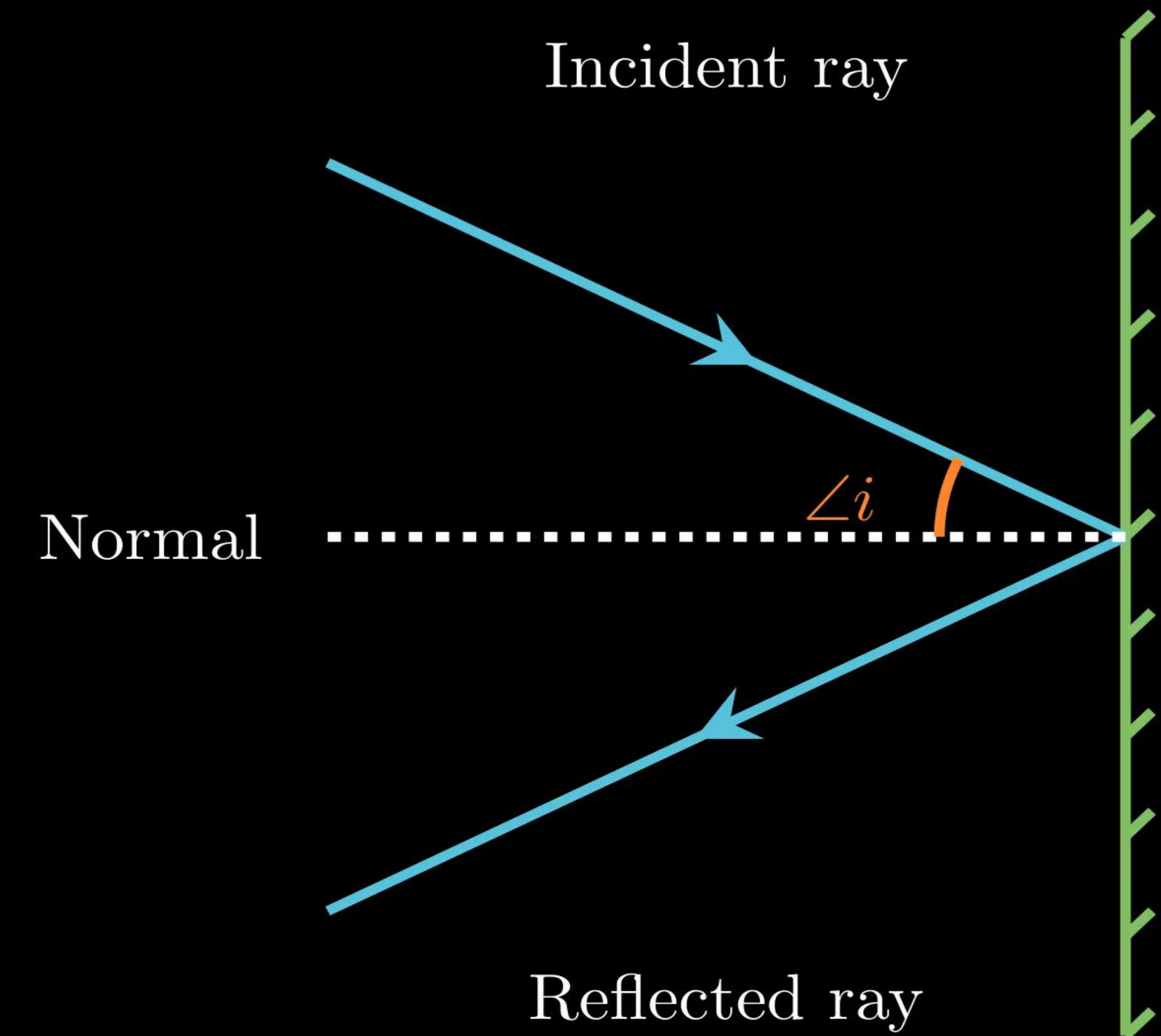
- The human eye is sensitive only to electromagnetic waves of wavelength about **400 nm to 750 nm**.
- Radiation in this range is called **light**.
- It is mainly through **light and the sense of vision** that we interpret the world around us.
- From daily experience, two basic facts about light are known:
 - 1. Light travels with **enormous speed**.
 - 2. Light travels in a **straight line**.
- It took time to realise that the speed of light is **finite and measurable**.
- The currently accepted value in vacuum is $c = 2.99792458 \times 10^8 \text{ m/s}$ ($3 \approx \times 10^8 \text{ m/s}$).

Introduction

- Though light is an **electromagnetic wave**, its **wavelength** is **very small** compared to everyday objects.
- Hence, in ordinary situations light travels in straight lines — called a **ray of light**.
- A group or bundle of rays forms a **beam of light**.
- In this chapter, we study the behaviour of light using the **ray model**.
- This model explains the phenomena of:
 - **Reflection** • **Refraction** • **Dispersion**
- Using the laws of reflection and refraction, we study **image formation by plane and spherical mirrors and lenses**. We also discuss important **optical instruments** such as the, **microscope**, and **telescope**

Reflection And Laws of Reflection

- **Reflection of Light:** When light is incident on a reflecting surface it bounces back into the same medium. This phenomenon is called reflection of light.
- **(a) Incident Ray:** Light ray which falls on the surface is called incident light.
- **(b) Reflected Ray:** Light ray which goes back after reflection is called reflected light.
- **(c) Normal:** An imaginary line drawn perpendicular to the reflecting surface.
- **(d) Angle of incidence ($\angle i$):** The angle between the incident ray and the normal.
- **(e) Angle of reflection ($\angle r$):** The angle between the reflected ray and the normal.



Reflection And Laws of Reflection

- **Laws of Reflection** : There are two laws of reflection:
- (i) The incident ray, the reflected ray and the normal at the point of incidence, all lie in the same plane.
- (ii) Angle of incidence is always equal to the angle of reflection i.e. $\angle i = \angle r$
- These laws of reflection are applicable to all types of reflecting surfaces including spherical surfaces.

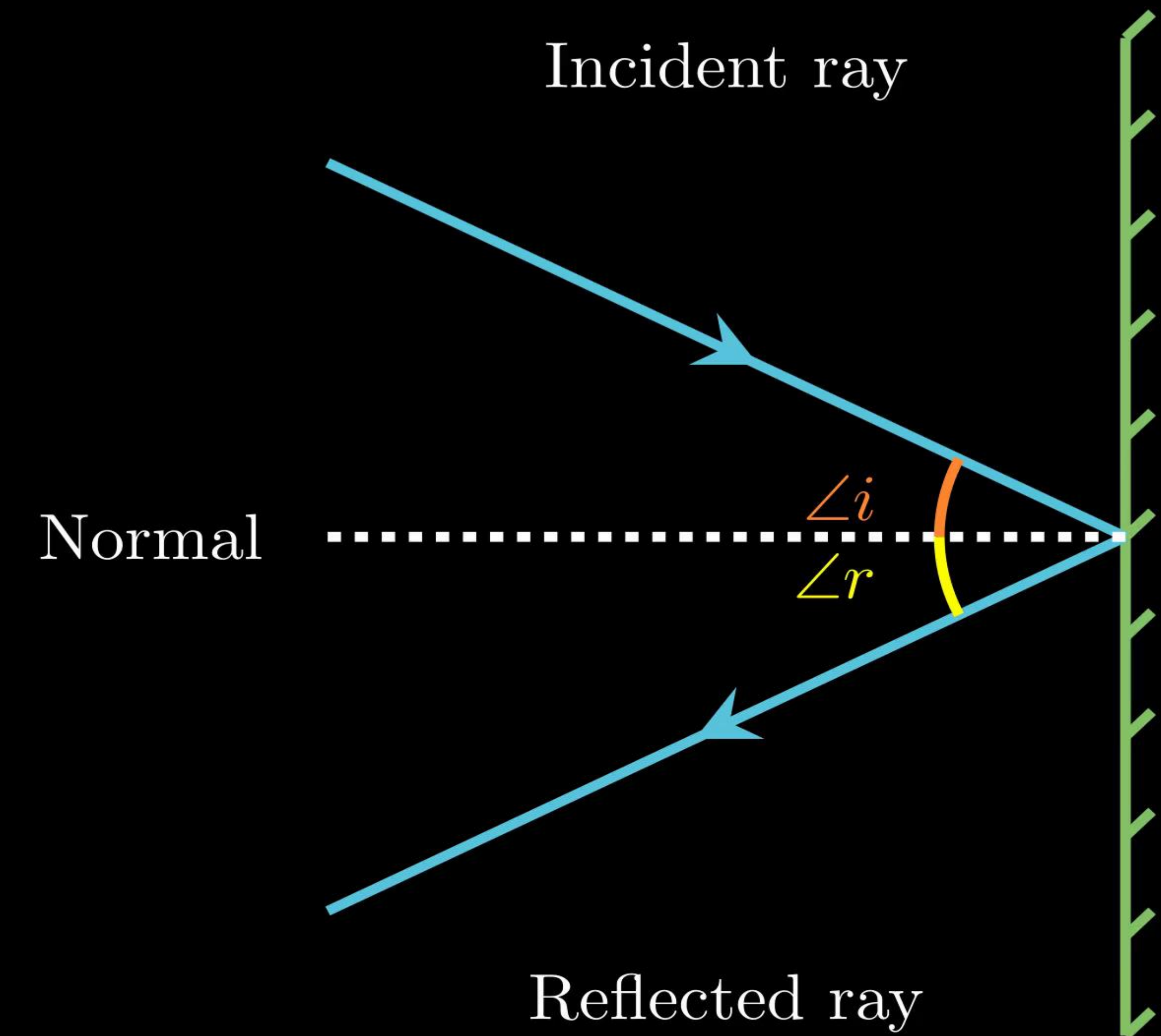
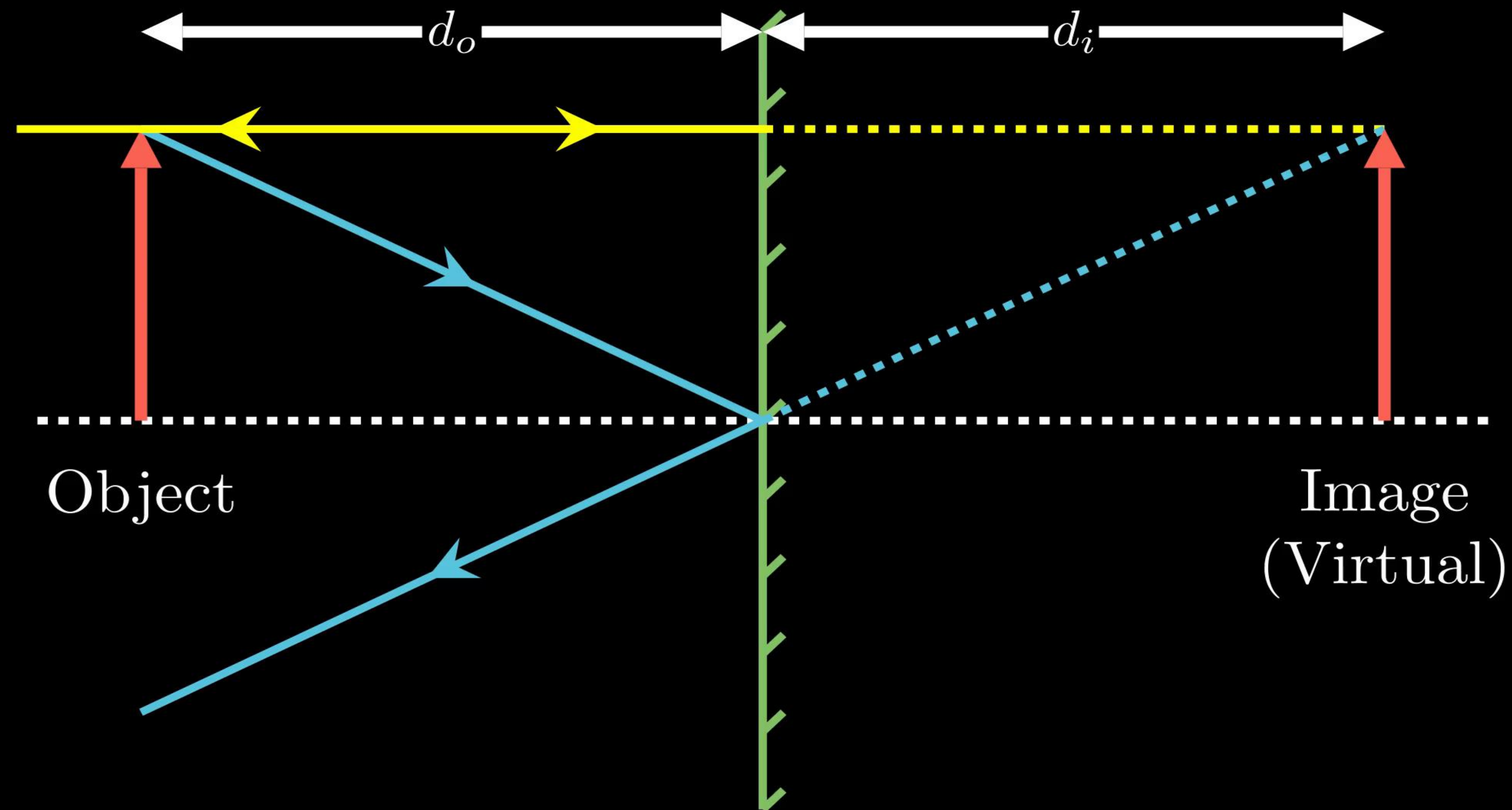


Image Formation by Plane Mirror:

- (i) Image formed by a plane mirror is always virtual and erect
- (ii) The size of the image is equal to that of the object.
- (iii) Distance of image from mirror = Distance of object from mirror
i.e., ($d_i = d_o$)
- (iv) The image is laterally inverted

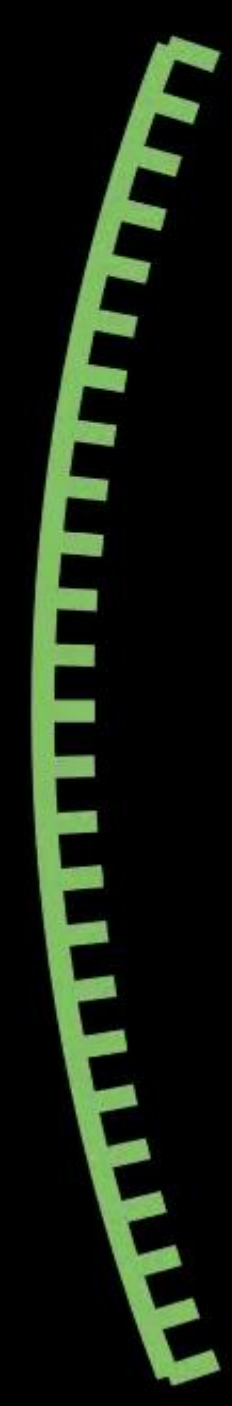


Spherical Mirrors

- **Spherical Mirrors** : Mirrors, whose reflecting surface can be considered to form a part of the surface of a sphere. The spherical mirror is of two types:
- (i) **Concave mirror** : A spherical mirror, whose reflecting surface is curved inwards, is called a concave mirror
- (ii) **Convex mirror** : A spherical mirror, whose reflecting surface is curved outwards, is called a convex mirror



Concave mirror

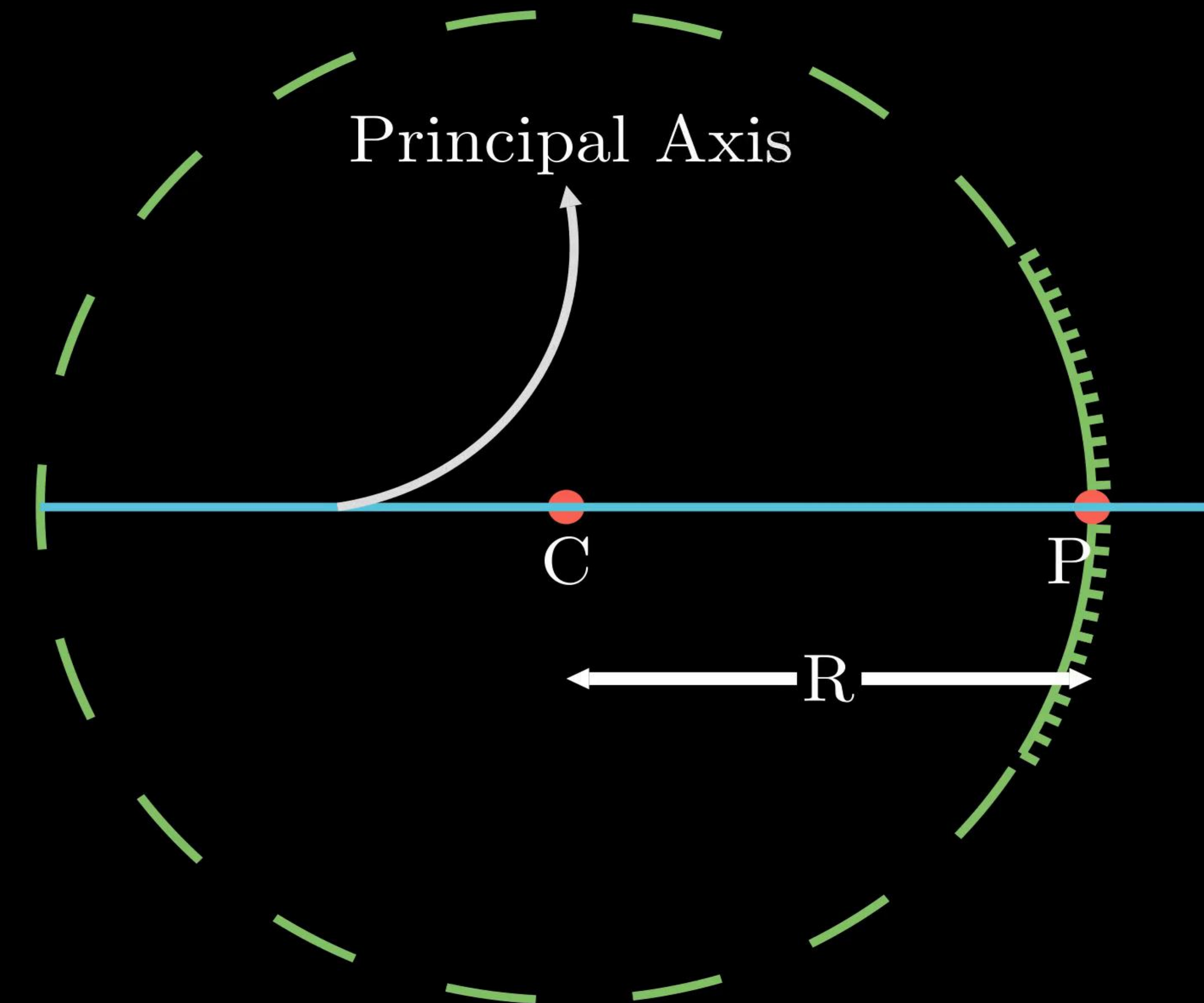


Convex mirror



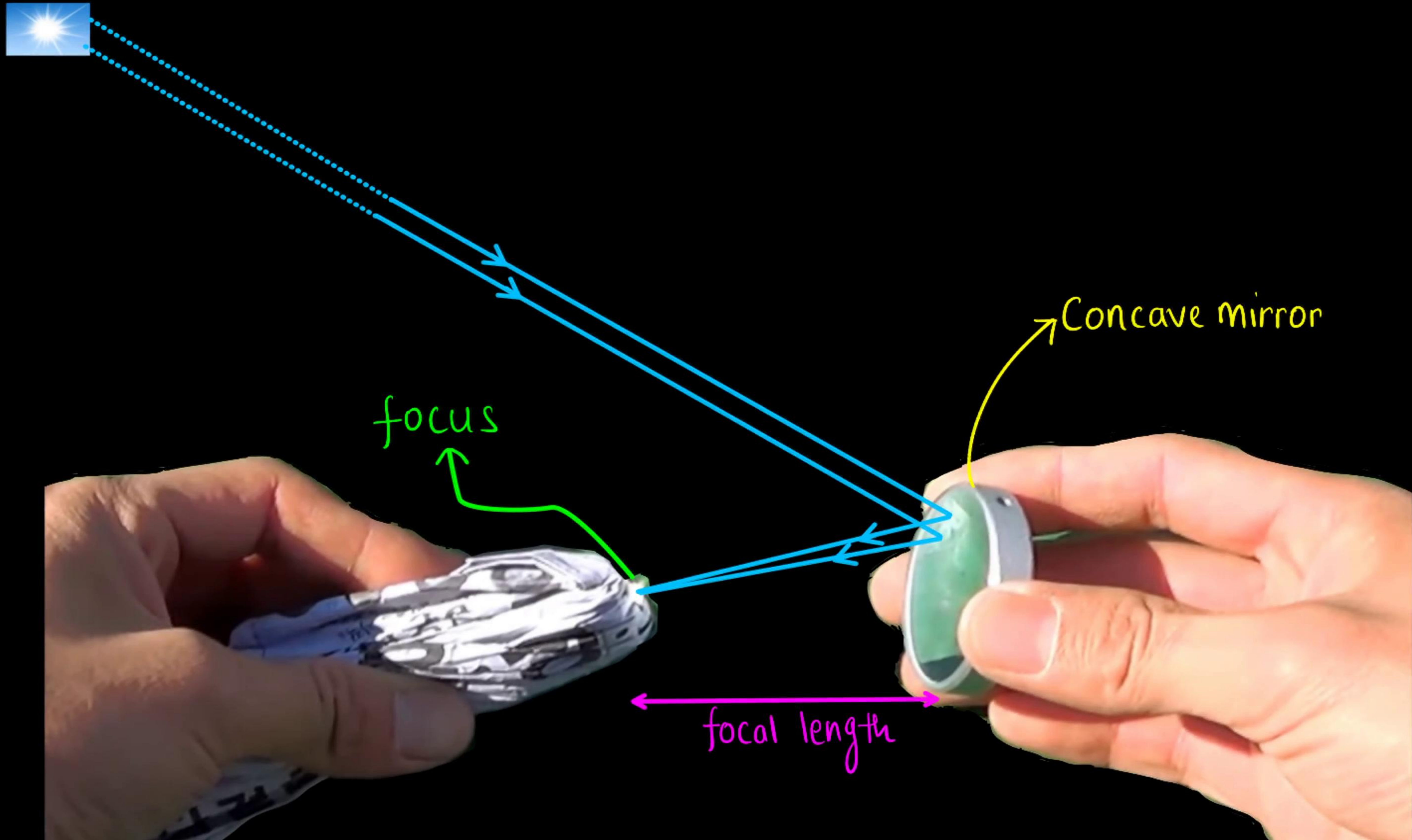
Some Definitions Related to Spherical Mirror:

- **Pole (P)**: The centre of the reflecting surface of a spherical mirror is called the pole. It lies on the surface of the mirror.
- **Centre of curvature (C)**: The centre of the sphere of which the mirror is a part is called the centre of curvature. Note that the centre of curvature is not a part of the mirror. It lies outside its reflecting surface
- **Radius of curvature (R)** : The radius of the sphere of which the mirror is a part is called the radius of curvature. Or the distance (PC) between pole and centre of curvature.
- **Principal axis** : An imaginary straight line passing through the pole and the centre of curvature of the mirror is called the principal axis.



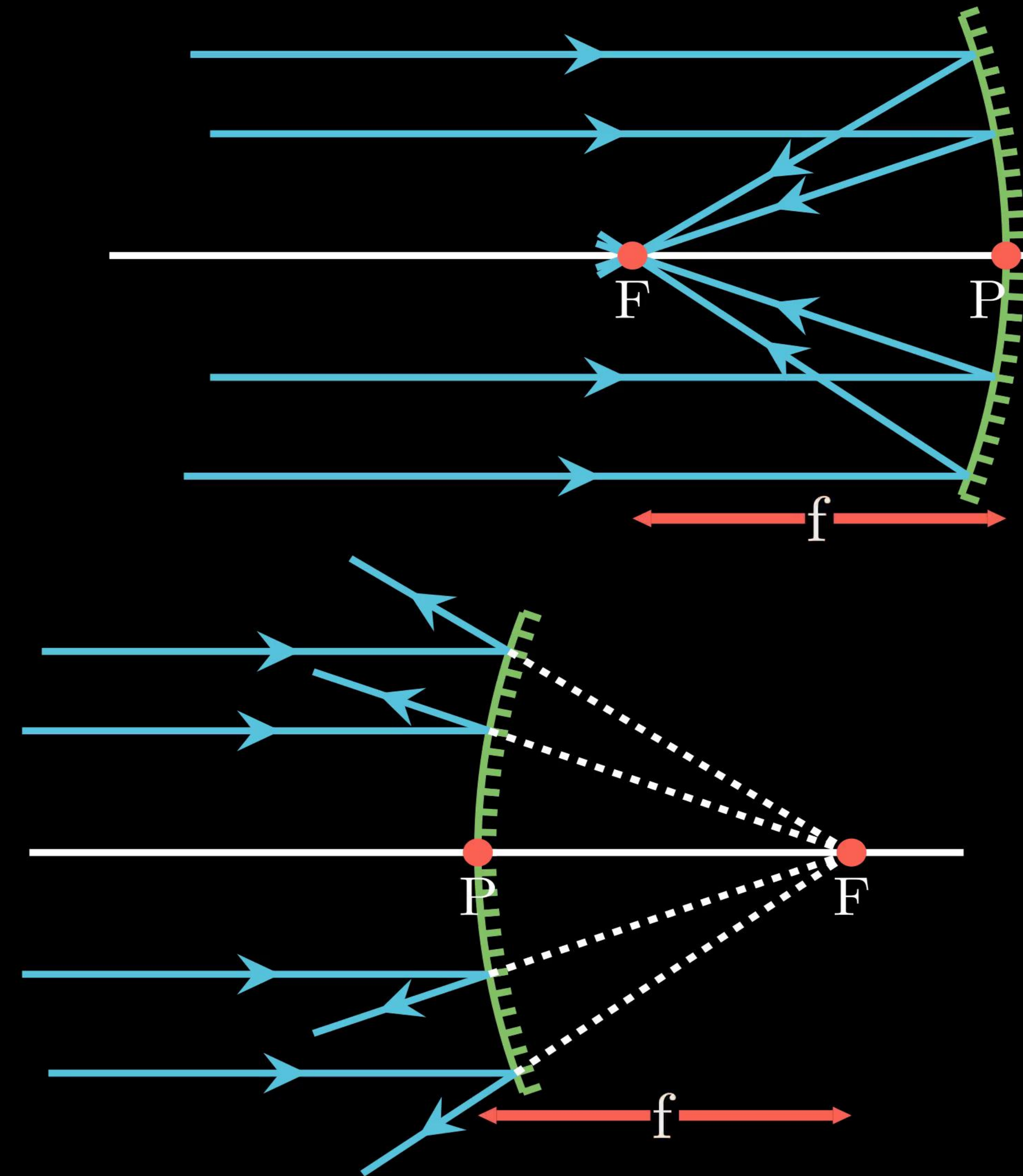
Some Definitions Related to Spherical Mirror:

Activity :



Some Definitions Related to Spherical Mirror:

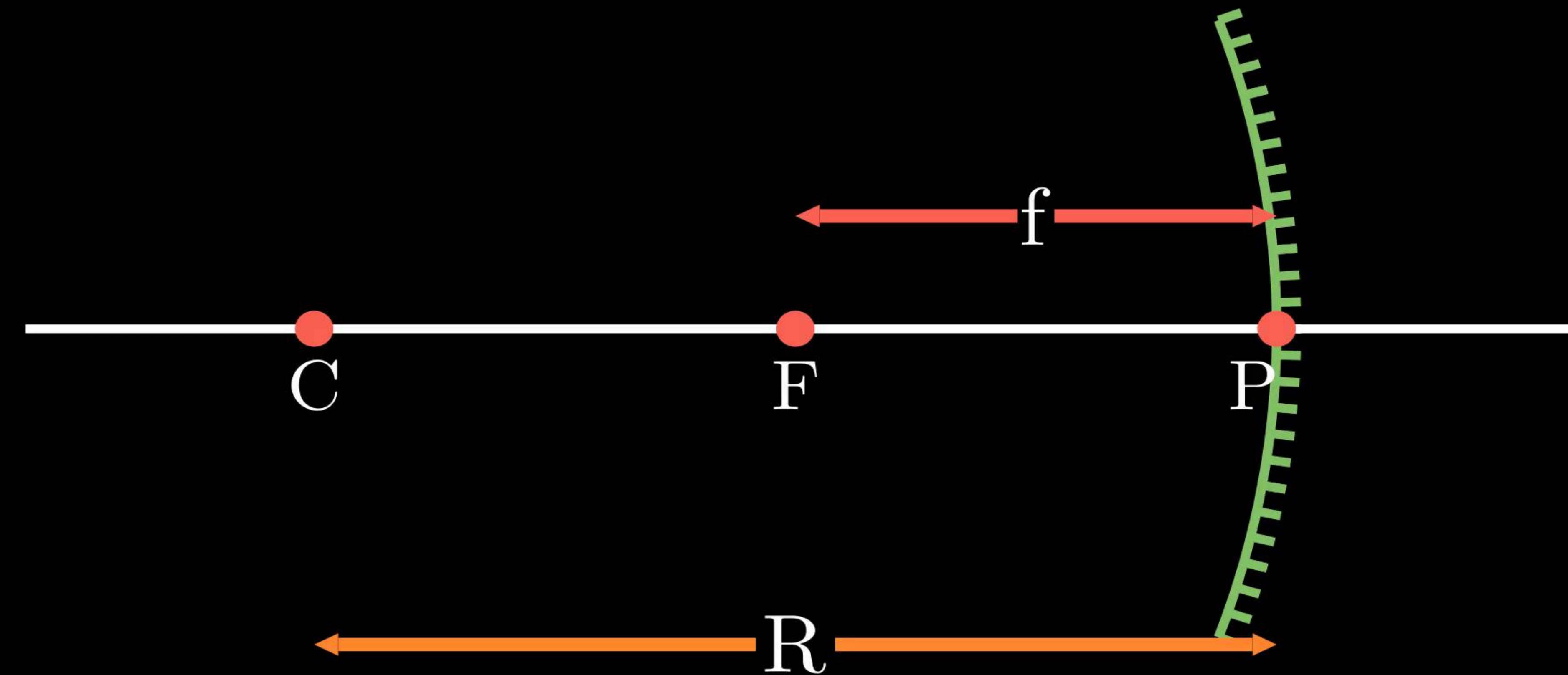
- **Principal Focus (F) of Concave Mirror:** When light rays parallel to principal axis incident on a concave mirror, after reflection they all meet at a point on the principal axis. This point is called principal focus (F) of concave mirror.
- **Principal Focus (F) of Convex Mirror:** When light rays parallel to principal axis incident on a convex mirror, after reflection they all appear to come from a point on the principal axis. This point is called principal focus (F) of convex mirror.
- **Focal Length (f) :** The distance between the pole (P) and the principal focus (F) is called the focal length



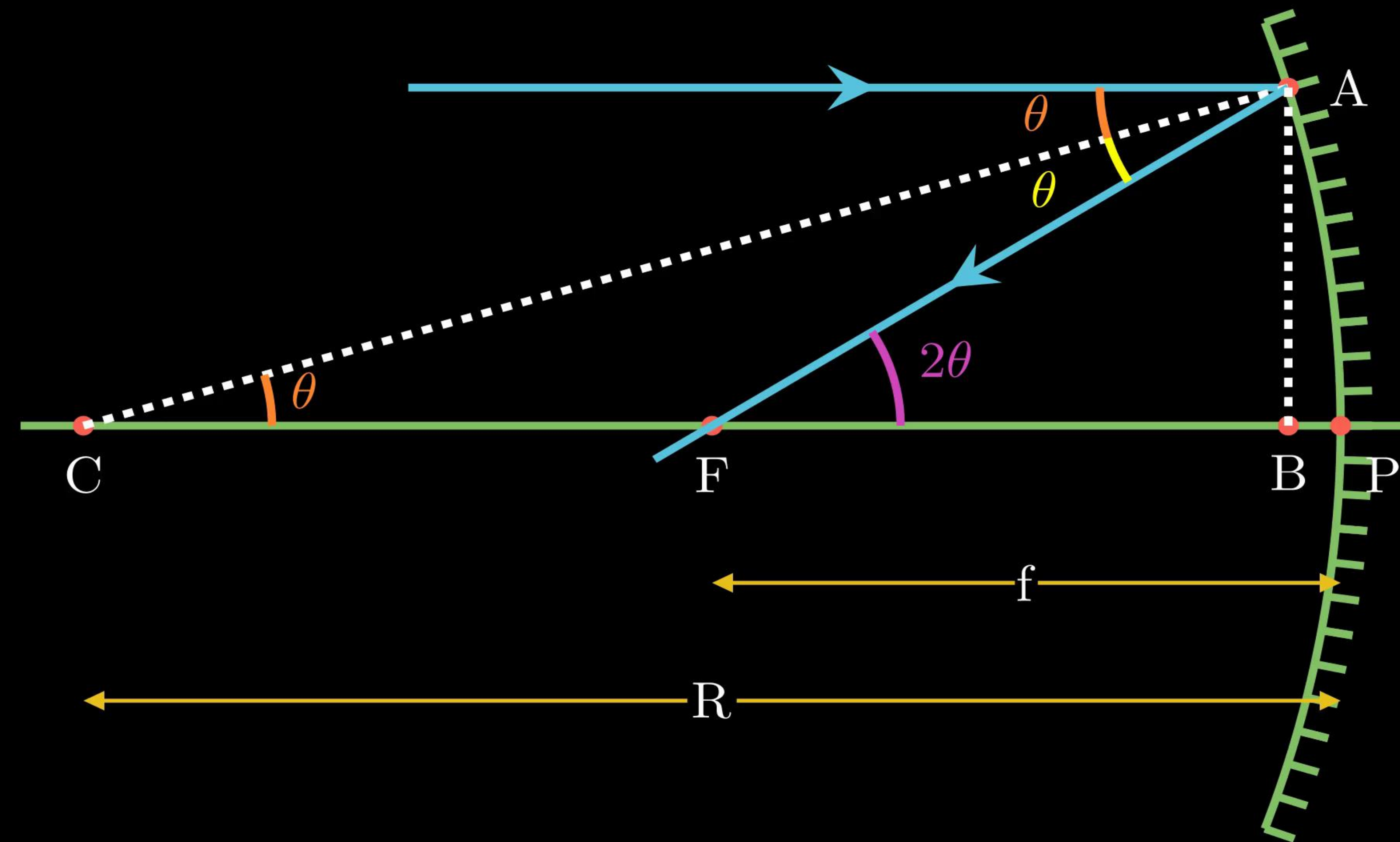
Some Definitions Related to Spherical Mirror:

- **Aperture** :The reflecting surface has a circular outline. The diameter of the reflecting surface of spherical mirror is called its aperture
- **Relation between the radius of curvature (R) and Focal Length(f)**:For spherical mirrors of small apertures, the radius of curvature is found to be equal to twice the focal length.

$$R = 2 \times f$$



Relation between Focal Length (f) and Radius of Curvature(R)



- In $\triangle ABC$ and $\triangle ABF$

$$\tan \theta = \frac{AB}{BC} \text{ and } \tan 2\theta = \frac{AB}{BF}$$

- For paraxial rays
 $\tan \theta \approx \theta$ and $\tan 2\theta \approx 2\theta$

$$\theta = \frac{AB}{BC} \text{ and } 2\theta = \frac{AB}{BF}$$

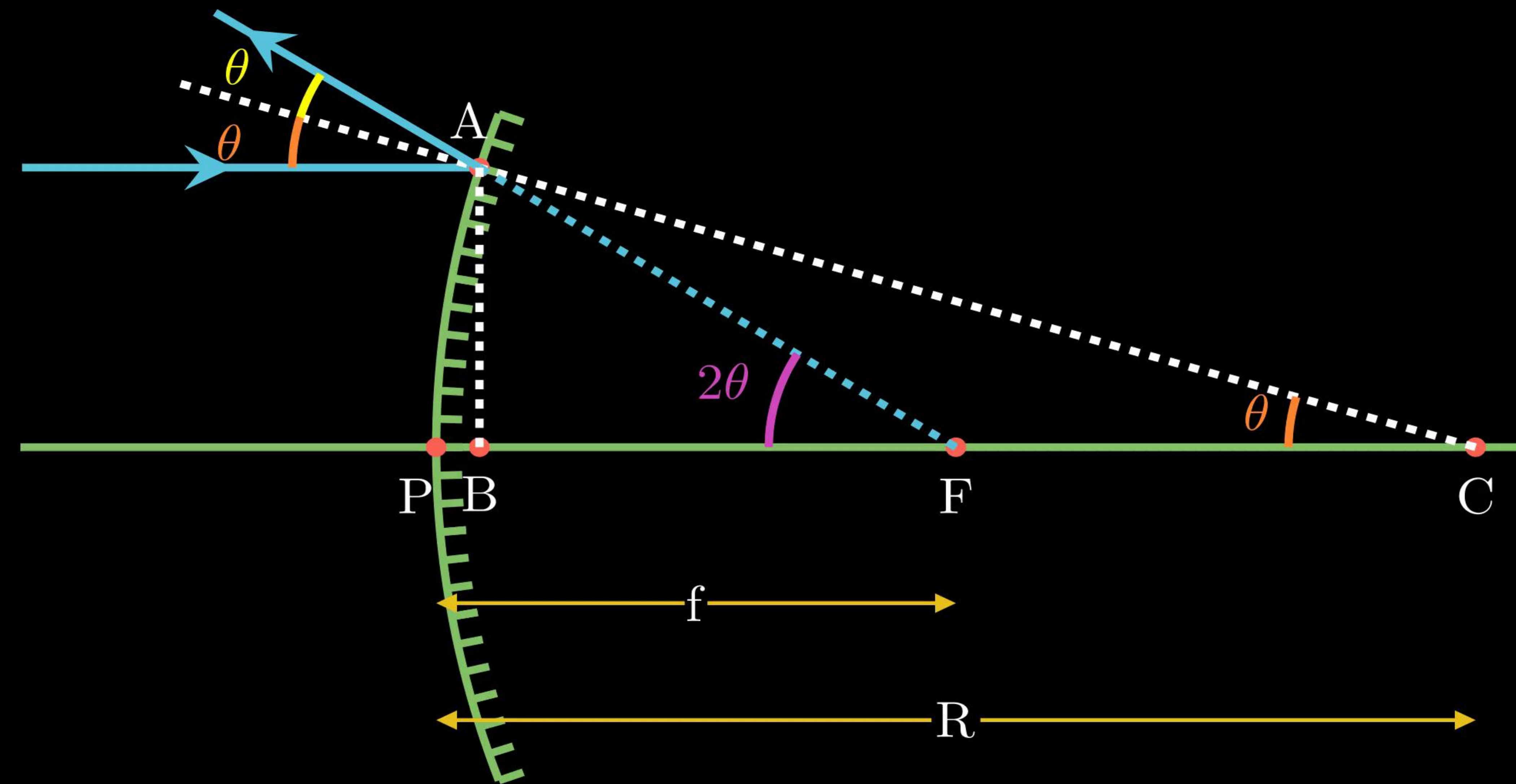
$$\frac{2AB}{BC} = \frac{AB}{BF}$$

$$2BF = BC$$

$$\therefore BF \approx PF = f \text{ and } BC \approx PC = R$$

$$\therefore 2f = R$$

Relation between Focal Length (f) and Radius of Curvature(R)



- In $\triangle ABC$ and $\triangle ABF$

$$\tan \theta = \frac{AB}{BC} \text{ and } \tan 2\theta = \frac{AB}{BF}$$

- For paraxial rays
 $\tan \theta \approx \theta$ and $\tan 2\theta \approx 2\theta$

$$\theta = \frac{AB}{BC} \text{ and } 2\theta = \frac{AB}{BF}$$

$$\frac{2AB}{BC} = \frac{AB}{BF}$$

$$2BF = BC$$

$$\therefore BF \approx PF = f \text{ and } BC \approx PC = R$$

$$\therefore 2f = R$$

Image formation by Spherical Mirrors

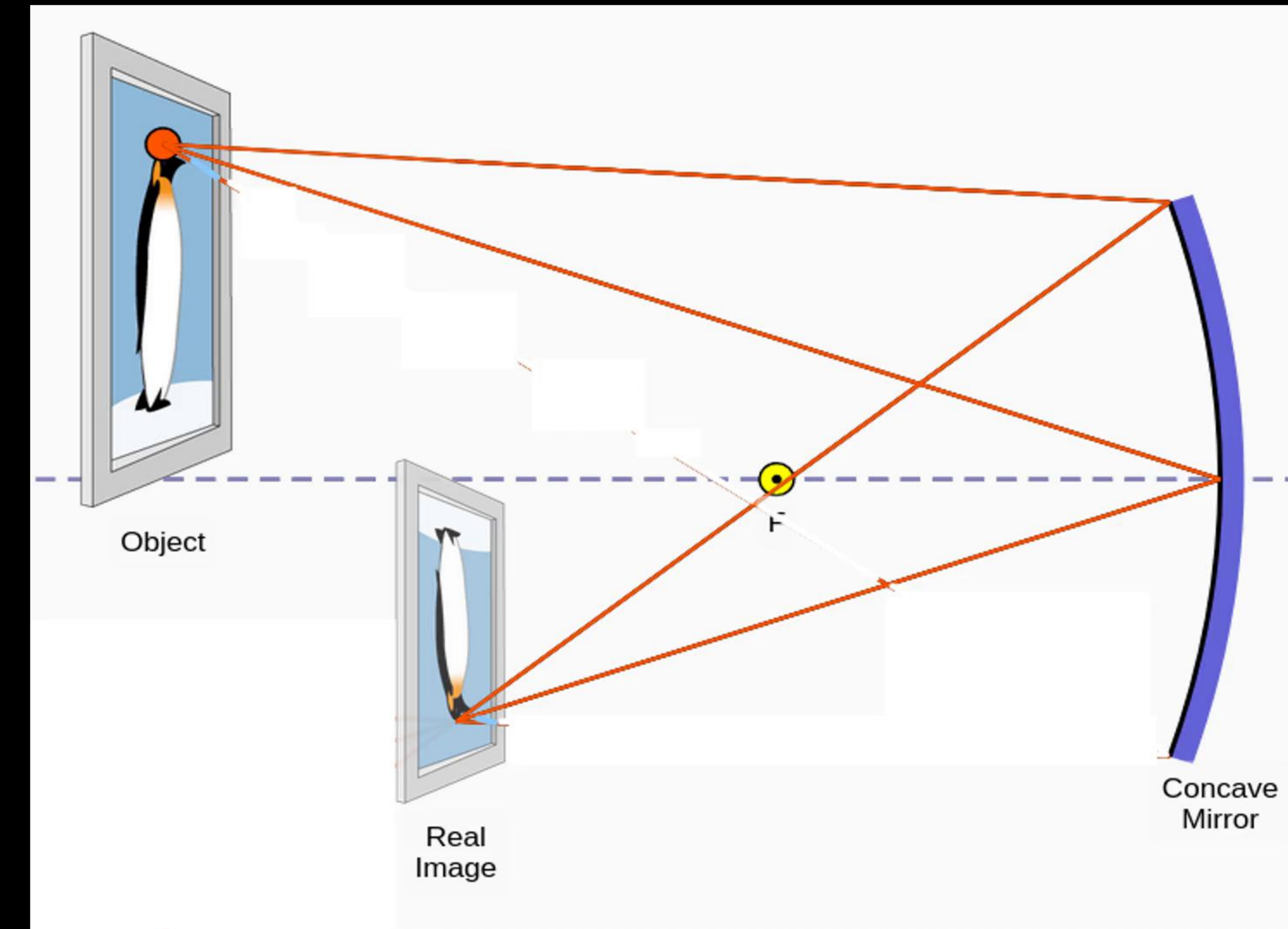
- **Image:** If light rays coming from a point after reflection meet at another point or appear to meet at another point, then second point is called image of the first point. There are two types of image, i.e.-
- **(a) Real image:** When the rays of light, after reflection from a mirror, actually meet at a point, then the image formed by these rays is said to be real. Real images can be obtained on a screen.
- **(b) Virtual image:** When the rays of light, after reflection from a mirror, appear to meet at a point, then the image formed by these rays is said to be virtual. Virtual images can't be obtained on a screen.

Image formation by Spherical Mirrors

	Position of the Object	Position of the Image	Size of the Image	Nature of the Image
(1)	At infinity	At the focus F	Highly diminished, point-sized	Real and inverted
(2)	Beyond C	Between F and C	Diminished	Real and inverted
(3)	At C	At C	Same size	Real and inverted
(4)	Between C and F	Beyond C	Enlarged	Real and inverted
(5)	At F	At infinity	Highly enlarged	Real and inverted
(6)	Between P and F	Behind the mirror	Enlarged	Virtual and erect

Representation of Images Formed by Spherical Mirrors Using Ray Diagrams

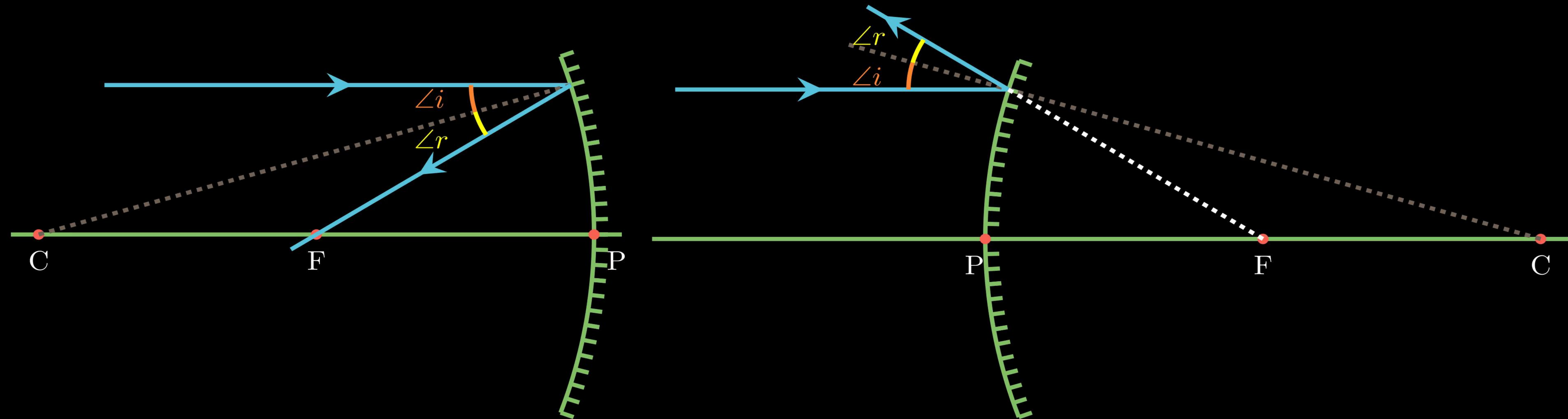
- Consider an extended object, of finite size, placed in front of a spherical mirror.
- Each small portion of the extended object acts like a point source.
- An infinite number of rays originate from each of these points.
- However, it is more convenient to consider only two rays, for the sake of clarity of the ray diagram.
- The intersection of at least two reflected rays give the position of image of the point object.



Representation of Images Formed by Spherical Mirrors Using Ray Diagrams

Any two of the following rays can be considered for locating the image.

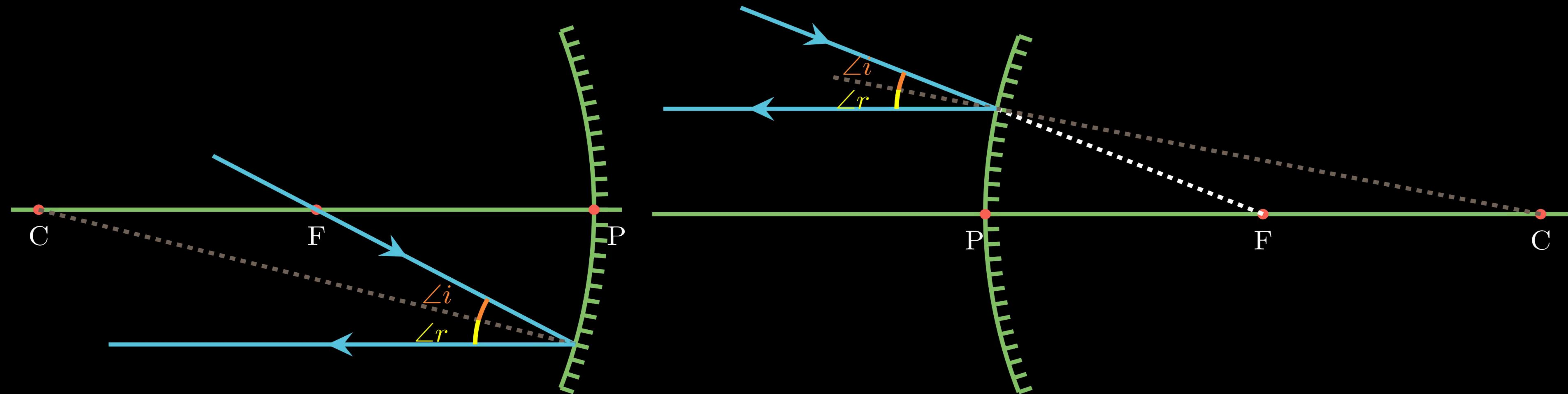
- (i) A ray of light which is parallel to the principal axis of a spherical mirror, after reflection, will pass through the principal focus in case of a concave mirror or appear to diverge from the principal focus in case of a convex mirror.



Representation of Images Formed by Spherical Mirrors Using Ray Diagrams

Any two of the following rays can be considered for locating the image.

- (ii) A ray passing through the principal focus of a concave mirror or a ray which is directed towards the principal focus of a convex mirror, after reflection, will emerge parallel to the principal axis.



Representation of Images Formed by Spherical Mirrors Using Ray Diagrams

Any two of the following rays can be considered for locating the image.

- (iii) A ray passing through the centre of curvature of a concave mirror or directed in the direction of the centre of curvature of a convex mirror, after reflection, is reflected back along the same path.

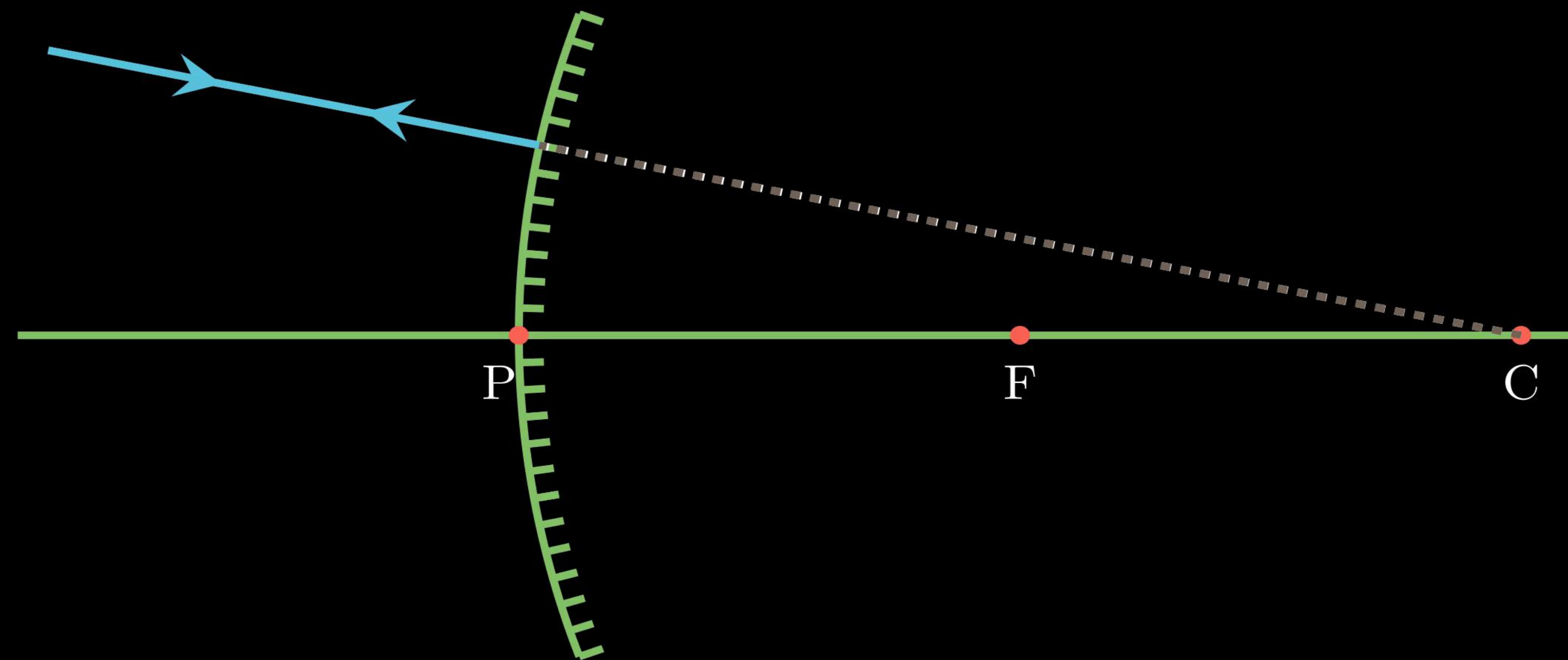
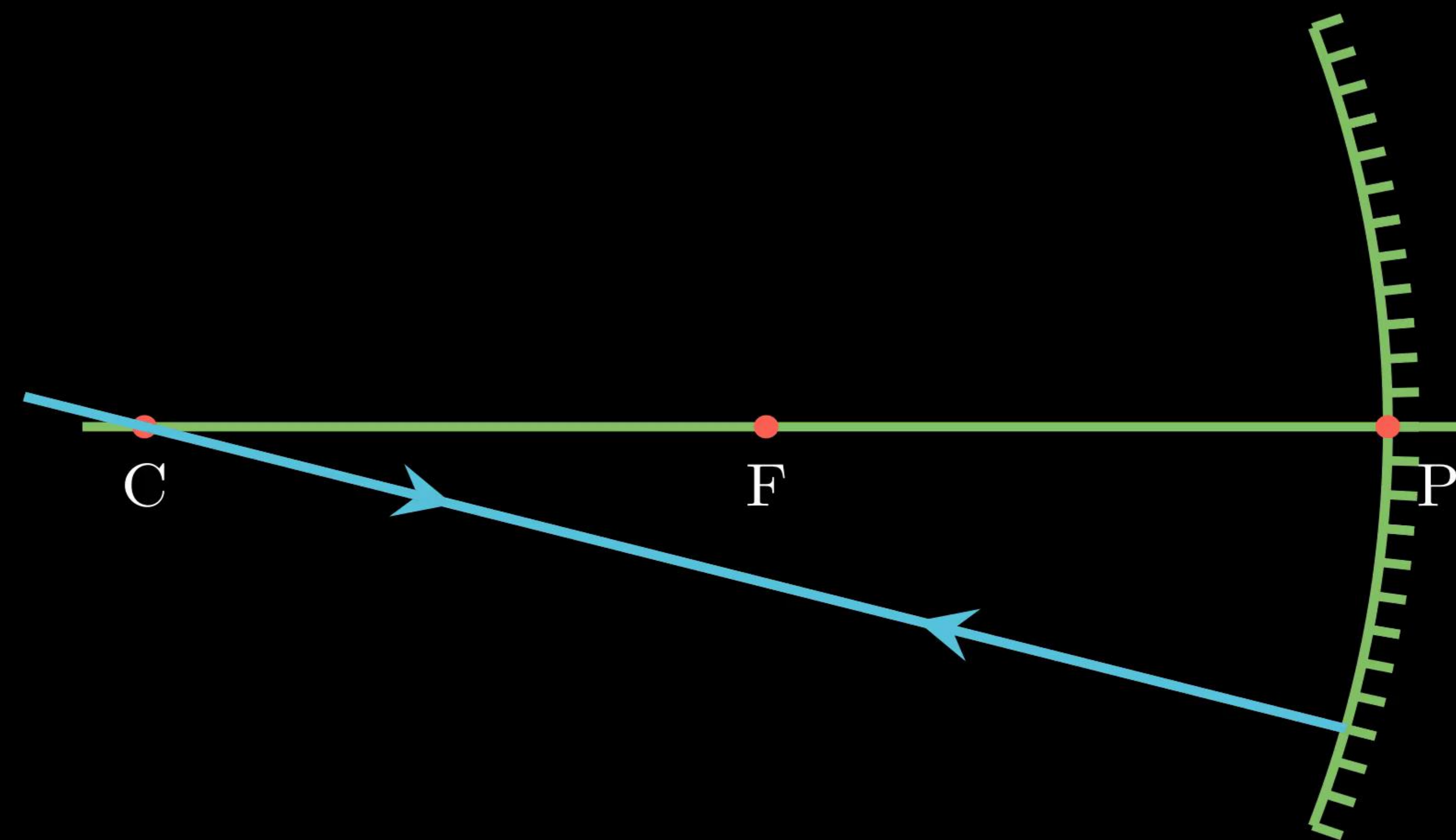
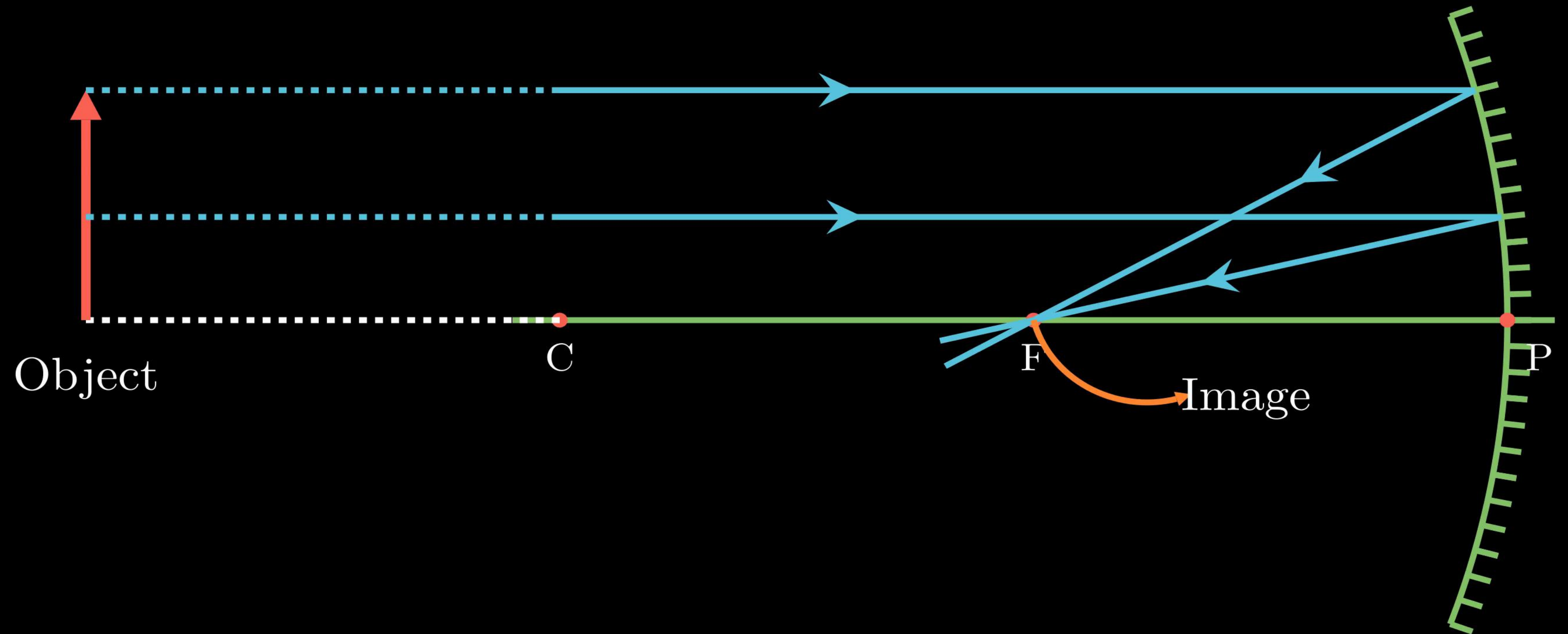


Image formation by Concave Mirror

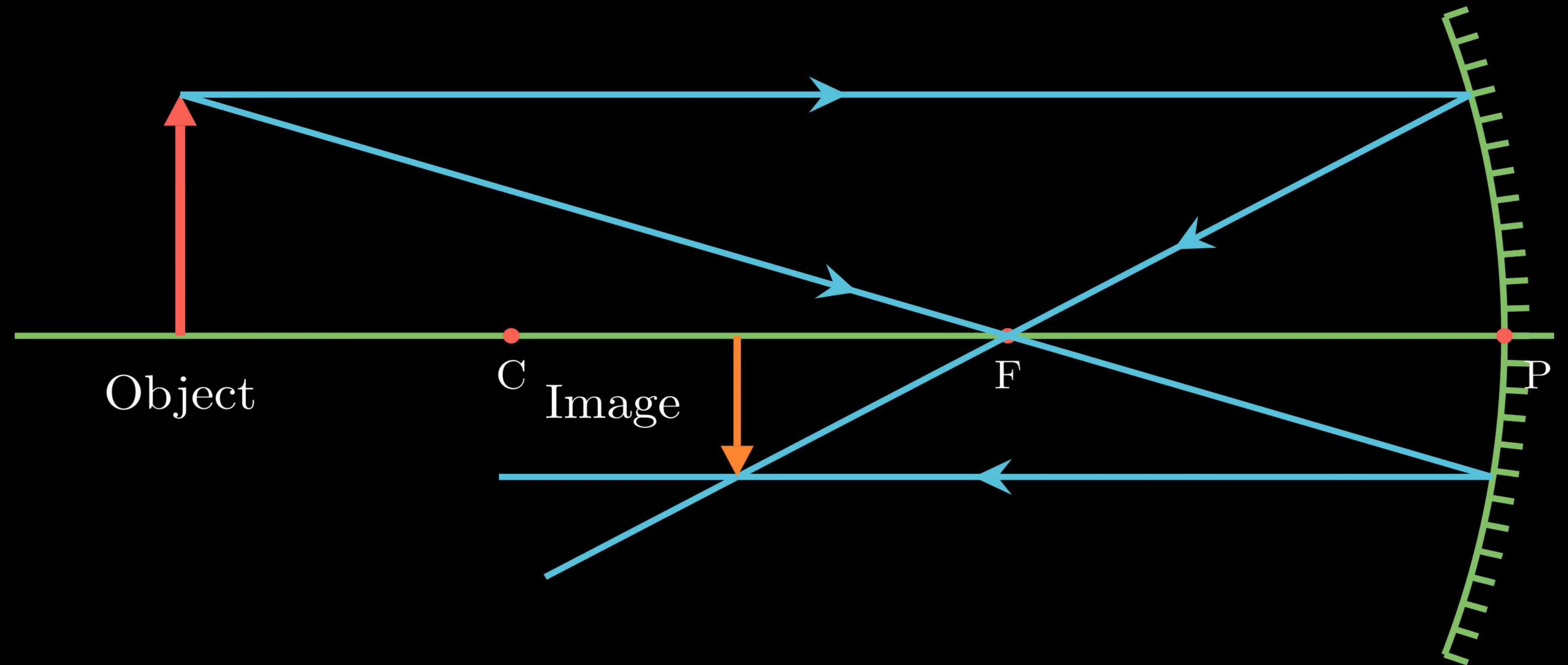
Position of Object (i): At infinity



Position of the Image	Size of the Image	Nature of the Image
At the focus F	Highly diminished, point-sized	Real and inverted

Image formation by Concave Mirror

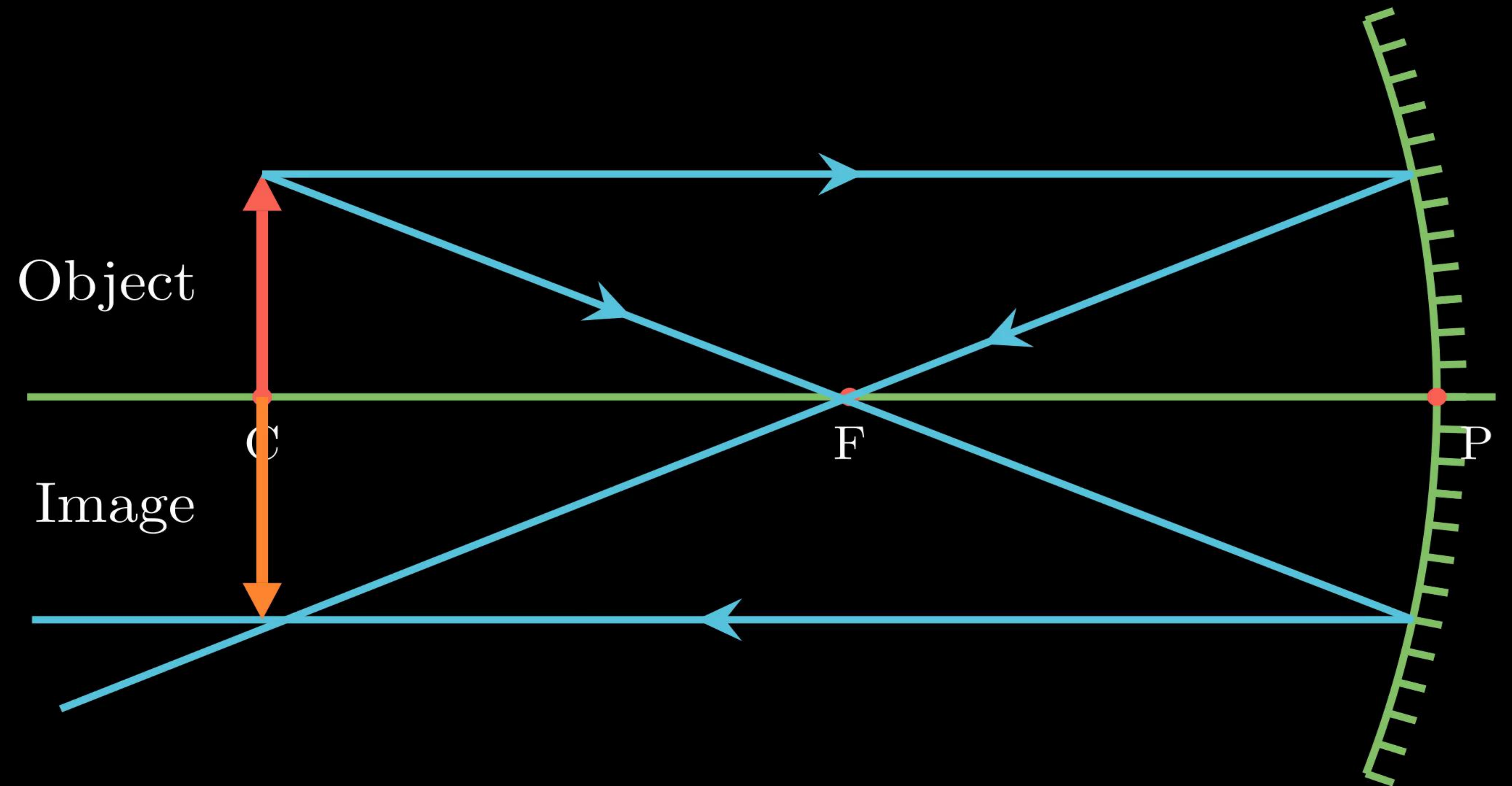
Position of Object (ii): Beyond C



Position of the Image	Size of the Image	Nature of the Image
Between F and C	Diminished	Real and inverted

Image formation by Concave Mirror

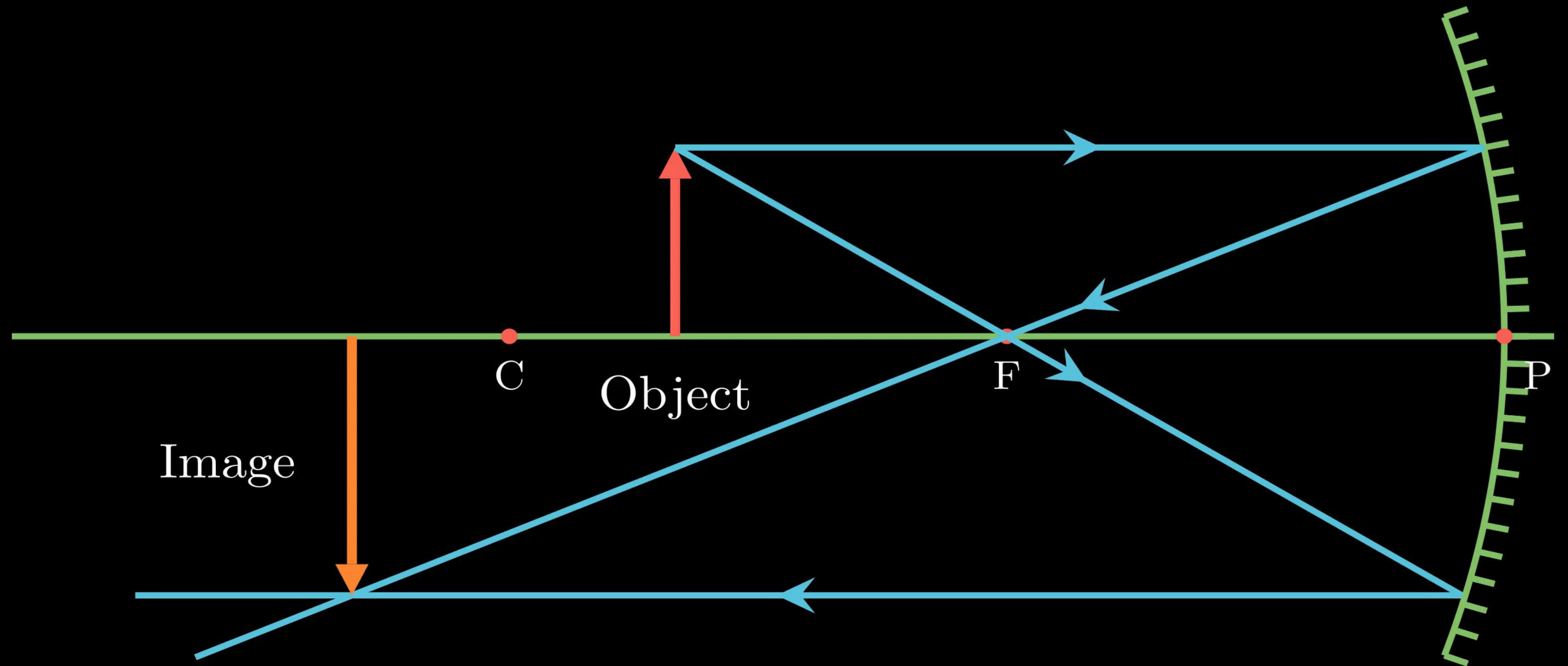
Position of Object (iii): **At C**



Position of the Image	Size of the Image	Nature of the Image
At C	Same size	Real and inverted

Image formation by Concave Mirror

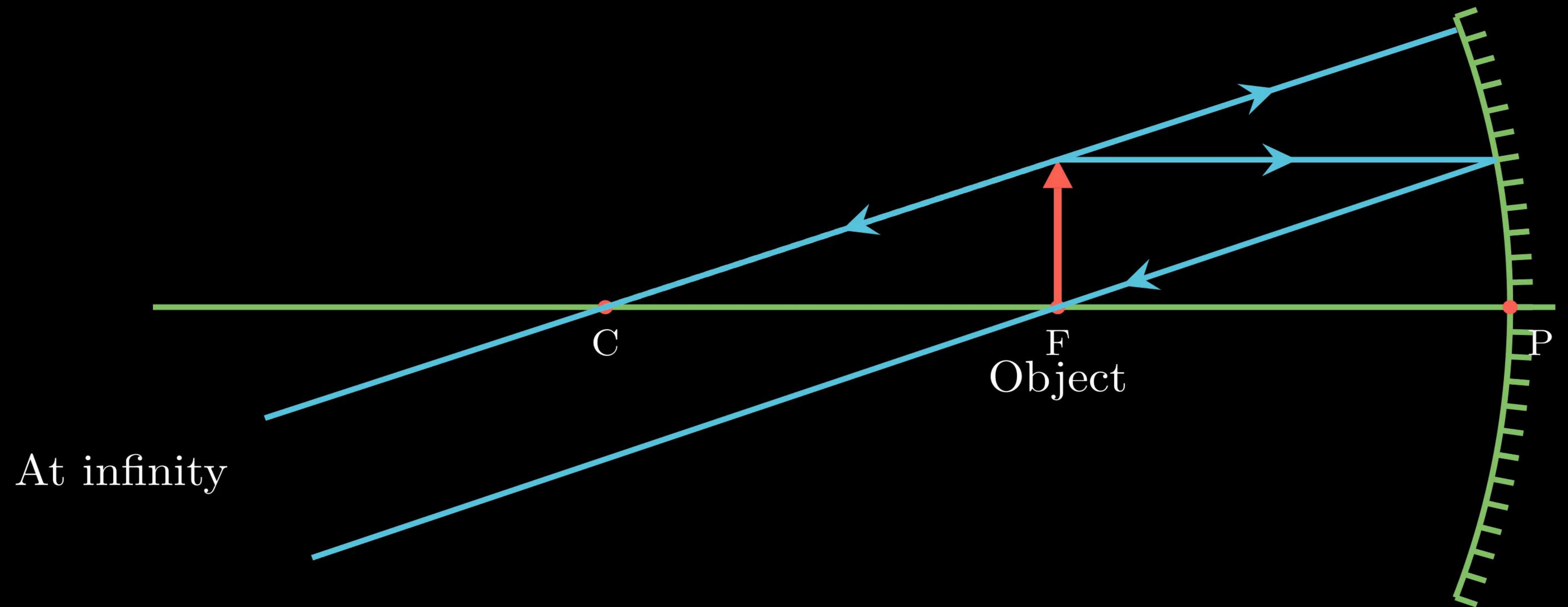
Position of Object (iv): **Between C and F**



Position of the Image	Size of the Image	Nature of the Image
Beyond C	Enlarged	Real and inverted

Image formation by Concave Mirror

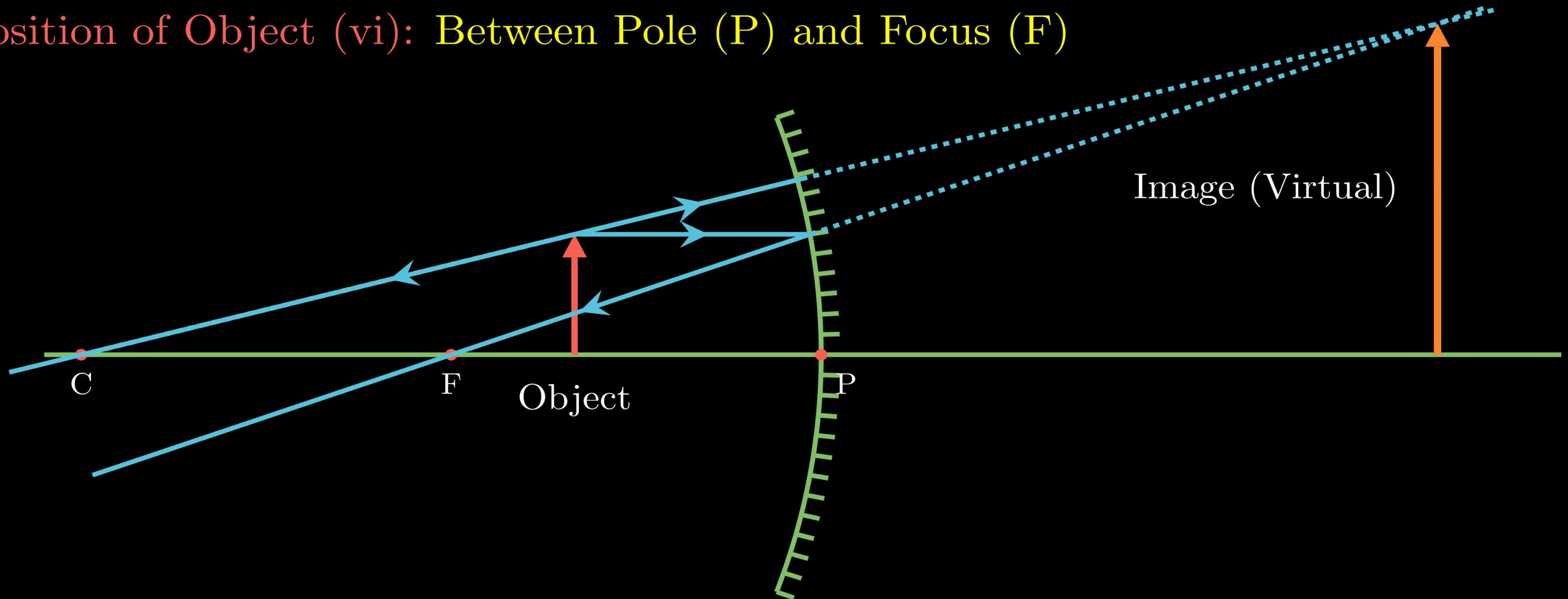
Position of Object (v): At F



Position of the Image	Size of the Image	Nature of the Image
At infinity	Highly enlarged	Real and inverted

Image formation by Concave Mirror

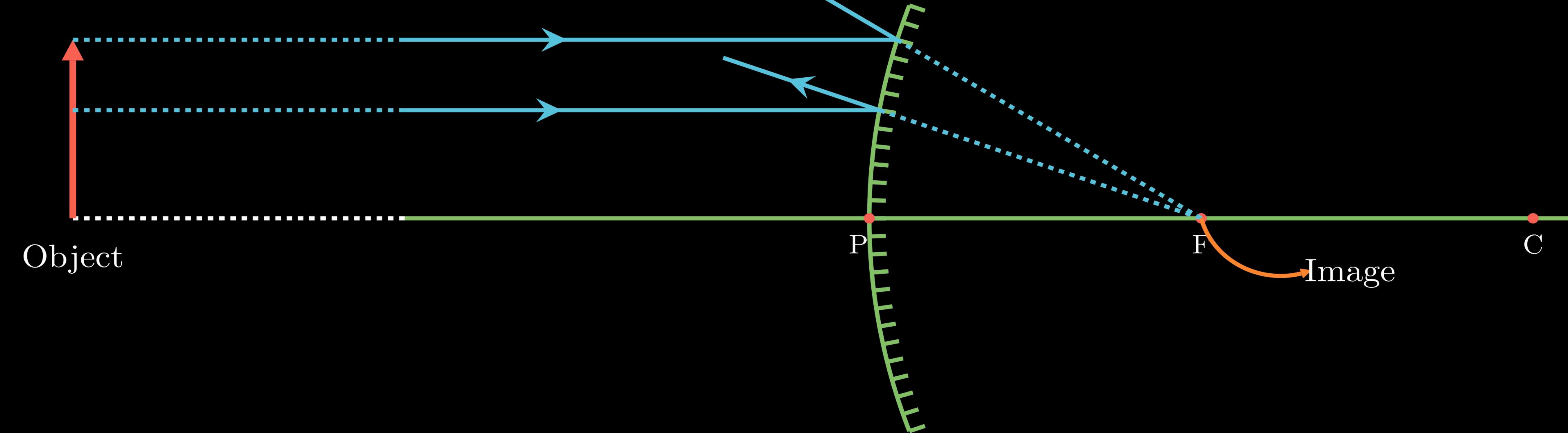
Position of Object (vi): Between Pole (P) and Focus (F)



Position of the Image	Size of the Image	Nature of the Image
Behind the mirror	Enlarged	Virtual and erect

Image formation by Convex Mirror

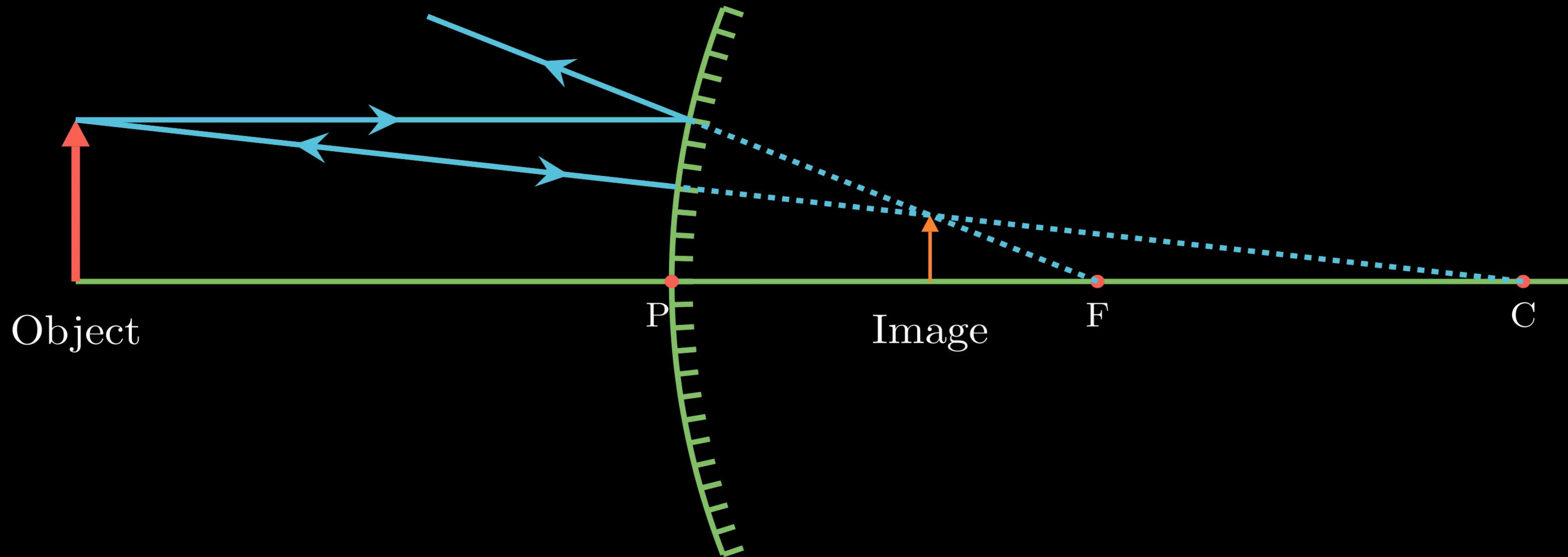
Position of Object (i): At infinity



Position of the Image	Size of the Image	Nature of the Image
At the focus F, behind the mirror	Highly diminished, point-sized	Virtual and erect

Image formation by Convex Mirror

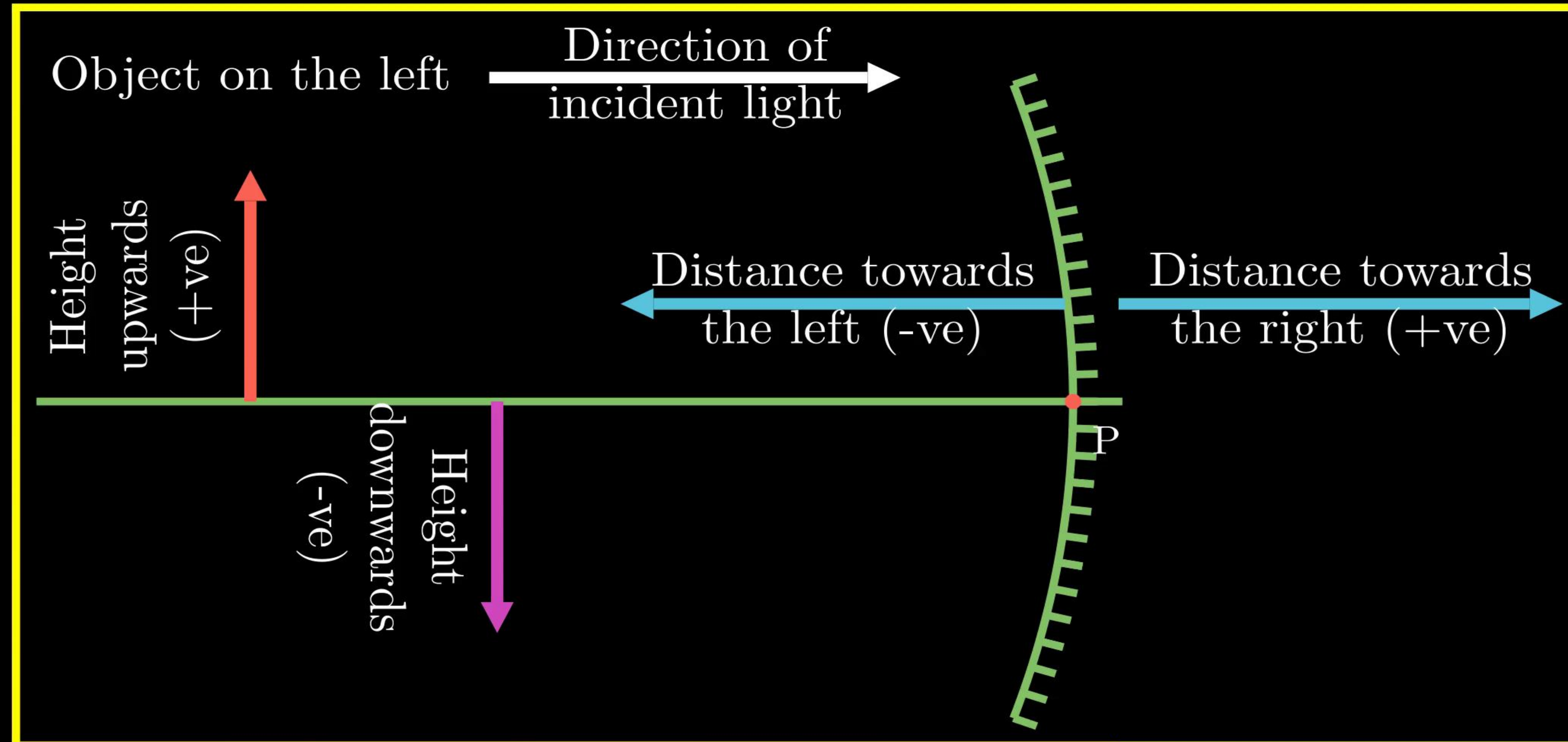
Position of Object (ii): Between infinity and the pole P of the mirror



Position of the Image	Size of the Image	Nature of the Image
Between P and F, behind the mirror	Diminished	Virtual and erect

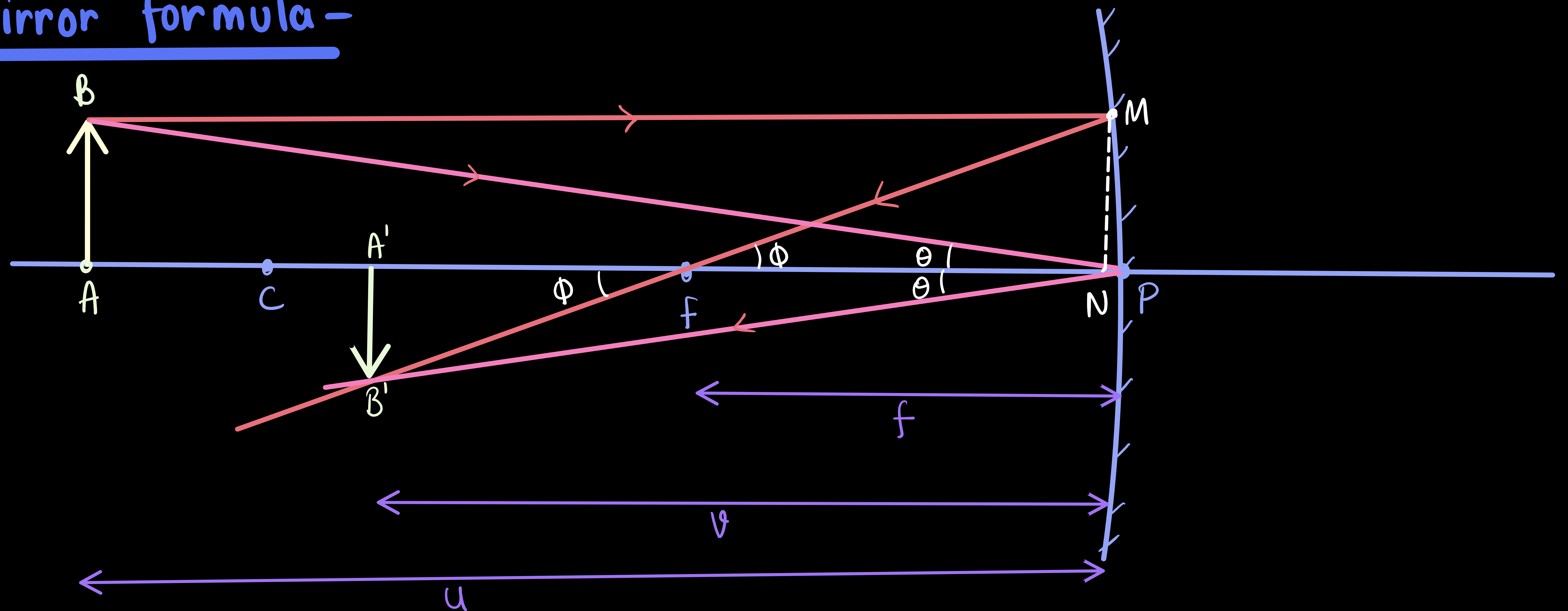
Sign Conventions For Spherical Mirror

- (i) All distances are measured from the pole of the mirror or the optical centre of the lens.
- (ii) All the distances measured in the same direction as the incident light (along $+x$ -axis) are taken as positive while those measured in the direction opposite to the direction of incident light are (along $-x$ -axis) are taken as negative.
- (iii) Distances measured perpendicular to and above the principal axis (along $+y$ -axis) are taken as positive.
- (iv) Distances measured perpendicular to and below the principal axis (along $-y$ -axis) are taken as negative.



Derivation - Mirror formula -

①



In $\Delta A'B'F$ and ΔMNF

$\angle B'A'F = \angle MNF$ and $\angle B'FA' = \angle MFN = \phi$ (V.O.A)

$\therefore \Delta B'A'F \sim \Delta MNF$ (By AA similarity rule)

$$\frac{B'A'}{MN} = \frac{A'F}{NF}$$

or $\frac{B'A'}{AB} = \frac{A'F}{PF}$ ① ($\because MN = AB$ and $NF \approx PF$)

In $\Delta A'B'P$ and ΔABP

$\angle B'A'P = \angle BAP = 90^\circ$ and $\angle BAP = \angle B'A'P = \theta$

$\therefore \Delta B'A'P \sim \Delta BAP$ (By AA similarity rule)

$$\frac{B'A'}{BA} = \frac{A'P}{AP}$$
 ②

from eq ① & ②

$$\frac{A'F}{PF} = \frac{A'P}{AP}$$

$$\frac{PA' - PF}{PF} = \frac{A'P}{AP}$$

$$\frac{PA'}{PF} - 1 = \frac{A'P}{AP}$$

$$\frac{-v}{-f} - 1 = \frac{v}{u}$$

$$+\frac{1}{f} - \frac{1}{v} = \frac{1}{u}$$

$$\boxed{\frac{1}{f} = \frac{1}{v} + \frac{1}{u}}$$

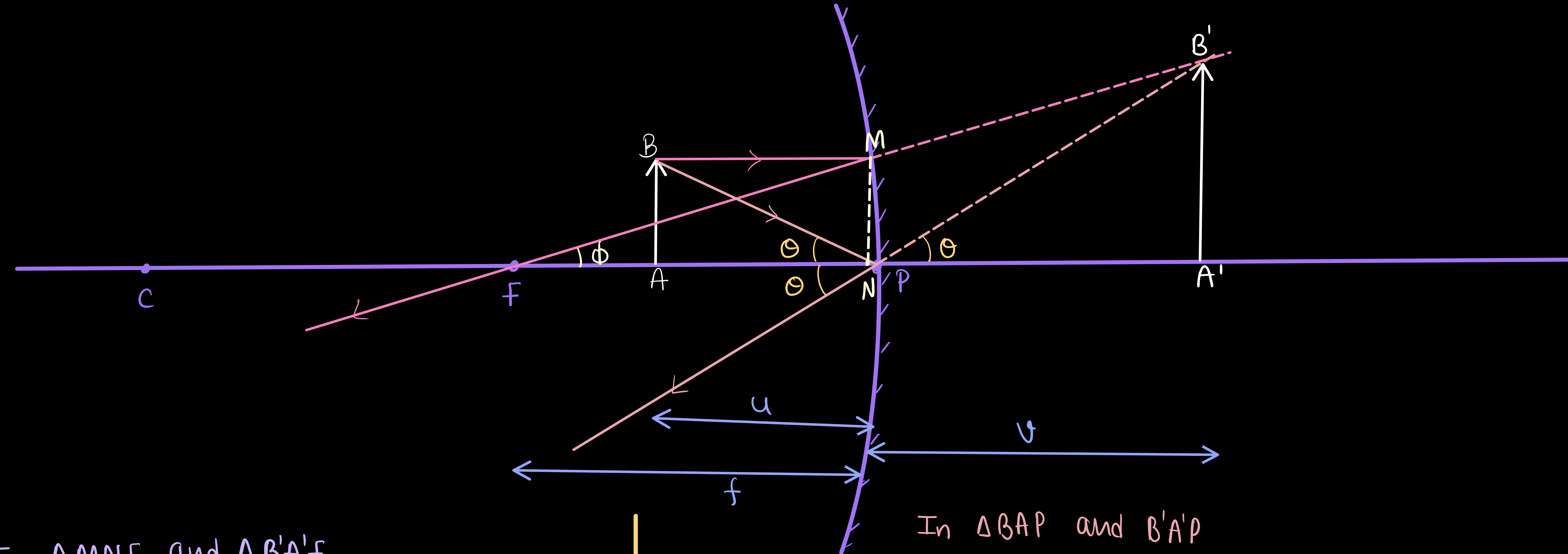
from eqⁿ (2)

$$\frac{B'A'}{BA} = \frac{A'P}{AP}$$

$$-\frac{h_i}{h_o} = \frac{-v}{u}$$

$$\boxed{m = \frac{h_i}{h_o} = \frac{-v}{u}}$$

(2)



In $\triangle MNF$ and $\triangle B'A'F$
 $\angle MNF = \angle B'A'F = 90^\circ$
 $\angle MFN = \angle B'FA' = \phi$
 $\therefore \triangle MNF \sim \triangle B'A'F$ (By AA ~)

$$\frac{MN}{B'A'} = \frac{NF}{A'F}$$
$$\frac{AB}{A'B'} = \frac{PF}{A'P + PF} \quad \text{--- (1) } \left(\because NF \approx PF \right. \\ \left. MN = AB \right)$$

In $\triangle BAP$ and $\triangle B'A'P$
 $\angle BAP = \angle B'A'P = 90^\circ$
 $\angle BPA = \angle B'PA' = \theta$
 $\therefore \triangle BAP \sim \triangle B'A'P$ (By AA ~)

$$\frac{AB}{A'B'} = \frac{AP}{A'P} \quad \text{--- (2)}$$

from eq (1) & (2)

$$\frac{PF}{A'P + PF} = \frac{AP}{A'P}$$

$$\frac{A'P + Pf}{Pf} = \frac{A'P}{AP}$$

$$\frac{A'P}{Pf} + 1 = \frac{A'P}{AP}$$

$$\frac{v}{-f} + 1 = \frac{v}{-u}$$

Dividing both sides by v

$$-\frac{1}{f} + \frac{1}{v} = -\frac{1}{u}$$

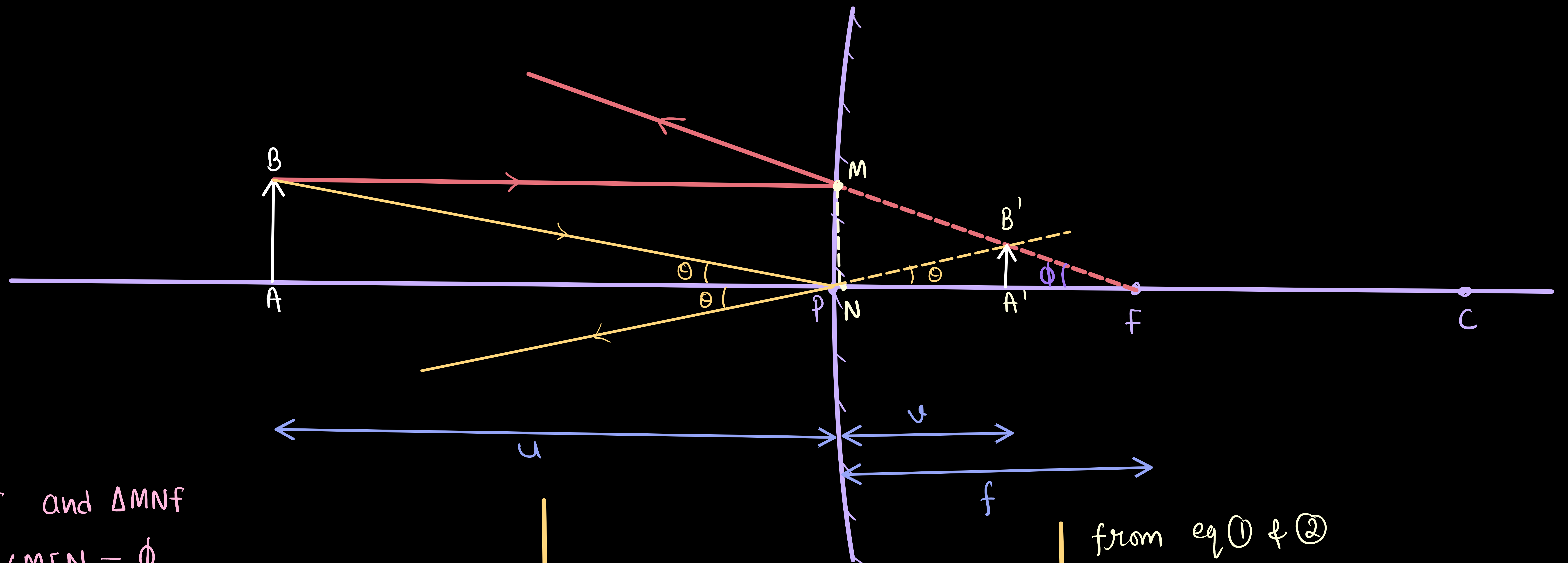
$$\boxed{\frac{1}{v} + \frac{1}{u} = \frac{1}{f}}$$

* Magnification (m)
from eq (2)

$$\frac{A'B'}{AB} = \frac{A'P}{AP}$$

$$\boxed{\frac{h_i}{h_o} = \frac{v}{-u}}$$

③



In $\Delta B'A'F$ and ΔMNF

$$\angle B'FA' = \angle MFN = \phi$$

$$\angle B'A'F = \angle MNF = 90^\circ$$

$$\therefore \Delta B'A'F \sim \Delta MNF$$

$$\frac{A'B'}{NM} = \frac{A'F}{NF}$$

$$\frac{A'B'}{AB} = \frac{PF - PA'}{PF} \quad \text{--- (1)} \quad \left(\because \begin{array}{l} NM = AB \\ NF \approx PF \end{array} \right)$$

In $\Delta A'B'P$ and ΔABP

$$\angle B'AP = \angle BAP = 90^\circ$$

$$\angle B'PA' = \angle BPA = \theta$$

$$\therefore \Delta A'B'P \sim \Delta ABP \text{ (By AA ~)}$$

$$\frac{A'B'}{AB} = \frac{A'P}{AP} \quad \text{--- (2)}$$

from eq (1) & (2)

$$\frac{PF - PA'}{PF} = \frac{A'P}{AP}$$

$$1 - \frac{PA'}{PF} = \frac{A'P}{AP}$$

$$1 - \frac{v}{f} = \frac{v}{-u}$$

Dividing by v

$$\frac{1}{v} - \frac{1}{f} = -\frac{1}{u}$$

$$\frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

* Magnification -

from eqⁿ (2)

$$\frac{A'B'}{AB} = \frac{A'P}{AP}$$

$$\frac{h_i}{h_o} = \frac{v}{-u}$$